BIOCOLOOP: COLLABORATIVE AGENTIC RESEARCH FOR BIOLOGICAL MODEL IMPROVEMENT

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ABSTRACT

Biological data are often distributed across laboratories that cannot pool their measurements, yet model development—choosing which configurations to train, evaluating them and deciding where to invest further training—is usually performed by a single party with centralized data. We introduce BioCoLoop, a framework that makes model development itself collaborative: laboratories fit and evaluate candidate configurations locally, and the resulting development evidence guides subsequent proposals and training allocation without pooling raw data. Task adapters support drug–target interactions, observed-response proteomic efficacy and cell perturbation identification. Following common federated-learning practice, we compare collaborative development with a laboratory working alone on common held-out tests. The same BioCoLoop workflow improves all five primary endpoints in mean when ten laboratories collaborate, raising DTI AUROC from 89.89 to 93.76 and proteomic AP from 20.13 to 36.99, and single-laboratory random, Bayesian and exhaustive search remain below the collaborative result. Training a fixed default design on all ten laboratories accounts for much of this gain; collaborative design search adds 1.96 AUROC on DTI and 6.77 Top-1 points on double-gene perturbation, where the default design degrades as laboratories are added. When laboratories hold distinct proteomic studies, collaboration raises target AUROC in seven of eight seeds (72.06 to 74.13), with seed-dependent AP gains; gains across distinct DTI and single-cell studies depend on source compatibility. Screening candidates on early collaborative evidence and training the promising ones longer improves double-gene perturbation identification by 10.84 percentage points over uniform allocation at matched training and evaluation budgets. In matched-access controls, language-model-guided search exceeds the exact random-search expectation in mean in all three task families, by task-dependent margins. These results establish a practical framework through which laboratories develop stronger biological predictors together.

1. INTRODUCTION

Artificial intelligence supports biological research through predictive modeling and the generation and evaluation of hypotheses (Cui et al., 2024; Theodoris et al., 2023; Roohani et al., 2024). Self-improving agents automate parts of model development (Jiang et al., 2025b; Lu et al., 2024; Zhou et al., 2026) through an iterative process of proposing, testing and revising (Madaan et al., 2023; Shinn et al., 2023; Huang et al., 2023). An agent proposes changes to a model design or its training hyperparameters, trains and evaluates the candidate, and uses the results to inform the next proposal.

In biology, useful data are often distributed across laboratories studying different cell types, perturbations or experimental assays. Data-sharing constraints can prevent these measurements from being pooled (Sheller et al., 2020; Bakhtiari et al., 2025; Dayan et al., 2021; Kaissis et al., 2020; Rieke et al., 2020). Collaborative learning enables laboratories to train a shared model through locally computed updates (McMahan et al., 2017; Wang et al., 2024; Li et al., 2020; Karimireddy et al., 2020; Kairouz et al., 2021). Automated federated design already combines configuration or system changes with distributed fitting (Khodak et al., 2021; Li et al., 2026). These systems design or tune the training procedure; the decisions of model development—which candidates to train, how to judge them and where to invest further training—still assume that one party can evaluate them. In biology, the evidence needed for these decisions is itself distributed across laboratories.

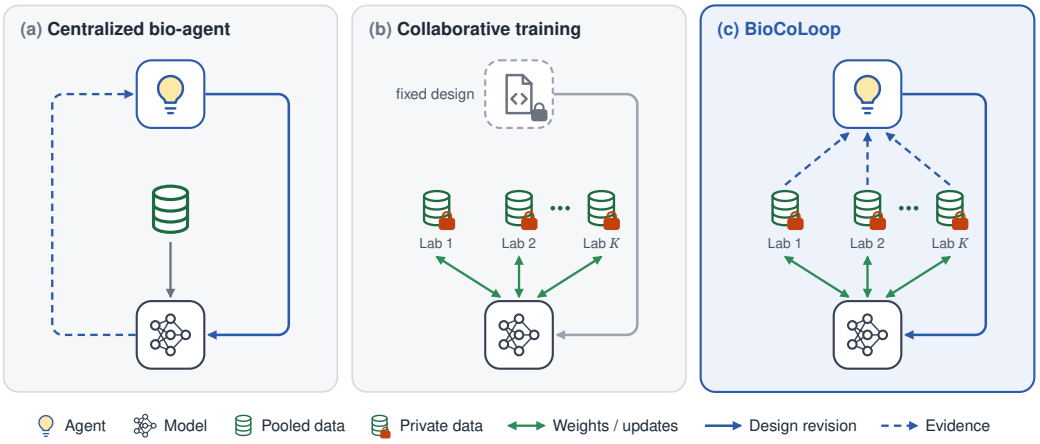


Figure 1: Research settings for biological AI. A centralized bio-agent revises a model using centrally accessible data. Collaborative training allows K laboratories to optimize the same model design and training hyperparameters by sharing model updates rather than raw data. BioCoLoop additionally uses results from local evaluation to guide subsequent changes to the model design and training hyperparameters.

Local development evidence provides a complementary research signal: it records how candidate configurations perform, how their training trajectories evolve and which trials succeed or fail. Sharing these summaries allows laboratories to inform subsequent experiments while retaining their measurements locally. *Can an iterative research process use laboratory-local training and evaluation to guide model development without pooling the underlying data?*

BioCoLoop extends collaboration from parameter fitting to model development by sharing both model updates and laboratory-local development evidence. Figure 1 places this framework alongside centralized self-improving agents and conventional collaborative training. An inner loop fits and evaluates candidates across laboratories. An outer loop uses the resulting evidence to revise proposals. Collaborative participation supplies training data and evaluation panels; development-history feedback uses their measured outcomes to guide search. We also evaluate a separate policy that uses early development trajectories to allocate further training.

We evaluate BioCoLoop on three biological task families with two proposal models, conventional search controls and task-adapted research-agent baselines. Following common practice in federated learning (Section 4.2), the central question is what laboratories gain by developing a model together rather than alone, measured on common held-out tests; same-access controls then characterize the search and allocation components. Our contributions are:

- We introduce BioCoLoop, a collaborative model-development framework in which laboratories jointly configure, train and evaluate biological predictors through a common interface while retaining raw measurements locally.
- We show that collaboration yields substantially stronger models: the same BioCoLoop workflow improves all five primary endpoints in mean with ten laboratories relative to one, beyond what single-laboratory random, Bayesian or exhaustive search attains, and laboratories holding distinct proteomic studies improve target ranking. A ten-laboratory fixed-design control separates the contribution of shared data from that of collaborative design search.
- We show that collaborative development evidence also guides training allocation at a matched budget, and use same-access controls to characterize how proposal history and promotion rules contribute.

2. Related work

Collaborative learning for biological data. Parameter aggregation enables institutions to train a shared model while retaining their measurements locally (McMahan et al., 2017; Sheller et al., 2020). Its benefit is conventionally measured by comparing collaborative training with each site training alone under a common held-out evaluation (Sheller et al., 2020; Chang et al., 2018; Dayan et al., 2021; Ogier du Terrail et al., 2022; Warnat-Herresthal et al., 2021; Saldanha et al., 2022), with clients constructed through synthetic dataset partitions (McMahan et al., 2017; He et al., 2020b) or natural entity partitions (Caldas et al., 2018); we adopt this protocol, using grouped synthetic partitions in the main study and study-defined laboratories in the source-transfer experiments. In single-cell analysis, scFed addresses cell-type classification and FedscGen addresses distributed batch correction (Wang et al., 2024; Bakhtiari et al., 2025). Tabula and its Chiron implementation support collaborative single-cell foundation-model training through an agent-enabled platform (Ding et al., 2026; Mechtel et al., 2026). Automated design can interact with local training: FedEx tunes hyperparameters while training shared weights (Khodak et al., 2021), FedNAS searches architectures over distributed data (He et al., 2020a), and federated tuning is evaluated on held-out data after distributed validation (Holly et al., 2022), with FedHPO-Bench providing a benchmark suite for such methods (Wang et al., 2023). FedEx and FedNAS optimize a predefined numerical search space with a fixed algorithm; BioCoLoop instead lets a language-model agent propose typed design changes with stated hypotheses, returns per-laboratory development evidence to that agent, and allocates training among its proposals, so that the same interface serves different task models and research agents.

Self-improving agents in biological research. Self-improving agents turn experiment outcomes into subsequent computational proposals. AIDE searches candidate programs through drafting, debugging and improvement (Jiang et al., 2025b), while the AI Scientist and AI-Scientist-v2 connect ideas to executed experiments, replanning, tree search and writing (Lu et al., 2024; Yamada et al., 2025). DrugEvolve combines research, implementation, evaluation and algorithm memory for drug-development tasks (Zhou et al., 2026; GAIR-NLP DrugEvolve contributors, 2026). BioCoLoop studies how multiple data-owning laboratories can supply development feedback through a shared research interface. TAPB, a ProteinTalks-derived efficacy head and scDEBART-based cell representations provide its task-specific prediction components (Lin et al., 2025; Sun et al., 2026; Sung & Kim, 2026). The proposal schema, fitting service and selection mechanism are common across tasks. Strong simple references on some perturbation benchmarks motivate evaluating the research procedure with fixed task predictors (Ahlmann-Eltze et al., 2025). Configuration search also connects to random search (Bergstra & Bengio, 2012), Bayesian optimization (Snoek et al., 2012) and language-model proposal distributions (Yang et al., 2024). We use feedback-free direct search to isolate the effect of development history under a shared proposal and fitting interface.

Collaborative scientific agents. The Virtual Biotech organizes specialist AI agents to integrate biomedical evidence for therapeutic discovery (Zhang et al., 2026). AutoScientists uses self-organizing agent teams with shared experimental state and recorded successes and failures (Gao et al., 2026). In both, the collaborators are agents sharing one data view. Helmsman, the closest precedent, uses LLM agents to generate and refine federated-learning systems in simulation (Li et al., 2026). In BioCoLoop the collaborators are data-holding laboratories and the research loop itself is federated: candidates are fitted and evaluated locally, and shared development evidence drives the next proposals and training allocation.

3. Method: collaborative biological model improvement

3.1 Problem setting and notation

Consider K laboratories working on the same biological prediction task. Laboratory i holds training data D_i^{train}, with $n_i = |D_i^{\text{train}}|$. Evaluation uses laboratory-local development panels $\{D_i^{\text{dev}}\}_{i\in\mathcal{P}}$, where $\mathcal{P}$ indexes the panels specified by the evaluation protocol. A task specification $\mathcal{T}$ defines the prediction inputs, outputs, training loss and evaluation metric. We write the task predictor as $f_{a,w}$, where $a \in \mathcal{A}$ specifies its design and w its trainable parameters. Given an initial design, a design space $\mathcal{A}$ and a research budget, BioCoLoop seeks a design and fitted predictor that perform well

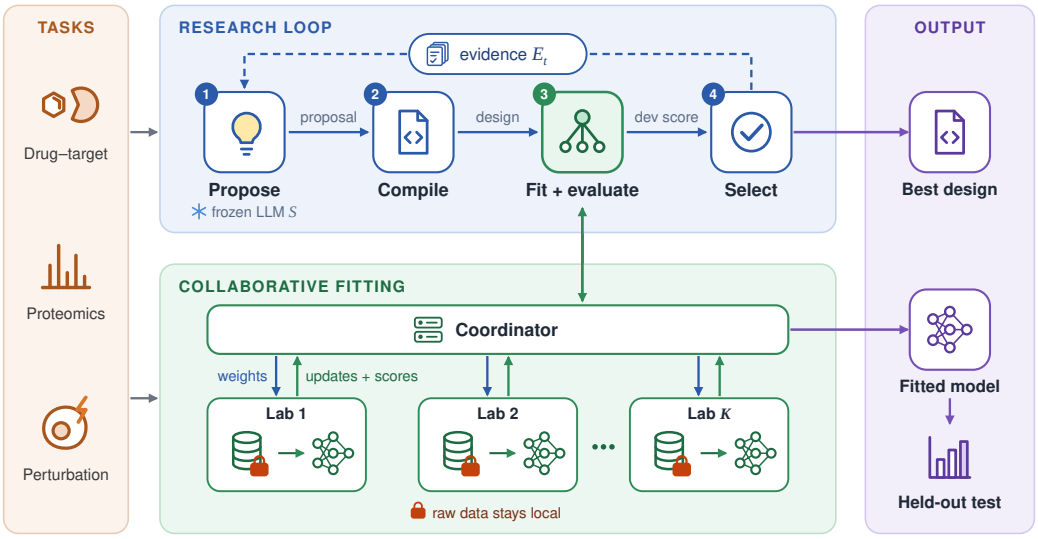


Figure 2: BioCoLoop's common research interface. A frozen research model S proposes a configuration, which is compiled into an executable design; the coordinator fits it with updates and development scores from K laboratories whose raw data stay local, and selection retains the best design. The ordered evidence history $\mathcal{E}_t$ conditions the next proposal, and the fitted model is then scored on the held-out test. Task adapters supply biological inputs, prediction heads and losses. Sections 3.2–3.3 define fitting, evidence and selection; Section 3.4 evaluates training allocation separately.

on the designated evaluation population. The three task families instantiate $f_{a,w}$ as task-specific predictors within this shared research procedure.

In the original end-to-end experiments, the *registered executable design space* $\mathcal{A}$ contains complete candidate configurations with four editable choices: learning rate, weight decay, coordinator momentum and an optional width-32 residual prediction module with zero-initialized output. Each `design_id` selects one configuration; the task adapter supplies the backbone and prediction head. The follow-up studies expand the head, objective and scheduling choices under separately fixed protocols (Appendices E.6 and E.9).

A *shared coordinator* is the trusted execution service used by all participating laboratories. It resolves each design identifier, initializes and broadcasts the predictor, coordinates local fitting, aggregates model states and bounded development measurements, and applies the checkpoint and design-selection rules. Laboratories receive the same candidate state and schedule while retaining raw training and development measurements locally. The coordinator supplies the aggregate evidence to research model S (Figure 2), a language model whose weights θ remain fixed throughout a run. We use this fitting interface to evaluate history-guided proposal revision (Section 3.3) and trajectory-guided training allocation (Section 3.4) separately.

3.2 Collaborative Fitting and Shared Evaluation

For each candidate a, the coordinator resets the predictor to its prescribed initialization w_0 and distributes it to all training laboratories. In the original protocol, $\mathrm{LocalEpoch}_a$ is one full local AdamW epoch with the candidate's learning rate and weight decay, batch size 64 and gradient-norm clipping at one. The optimizer state resets at each round. With sample weights $\pi_i = n_i / \sum_{j=1}^{K} n_j$, coordinator momentum coefficient β_a and initial velocity $v_0 = 0$, round r updates the trainable

parameters by

$$u_{i,r} = \mathrm{LocalEpoch}_a(w_{r-1}; D_i^{\mathrm{train}}), \quad \bar{w}_r = \sum_{i=1}^{K} \pi_i u_{i,r}, \\ v_r = \beta_a v_{r-1} + \bar{w}_r - w_{r-1}, \qquad w_r = w_{r-1} + v_r. \tag{1}$$

The coordinator carries momentum across aggregation rounds through v_r. With $\beta_a = 0$, aggregation reduces to sample-weighted parameter averaging (McMahan et al., 2017). In each round, the coordinator sends the current trainable state and fixed candidate settings to every laboratory. Laboratory i returns its locally fitted state $u_{i,r}$ and training-example count n_i. Frozen feature extractors remain unchanged. Task adapters supply the biological inputs, trainable prediction components and local loss; raw examples, expression profiles, labels and per-example identifiers remain local.

Every five rounds and at the final round, laboratories evaluate the aggregated predictor f_{a,w_r} on the development panels indexed by $\mathcal{P}$. Each returns a scalar primary score $s_i(a,r)$, development loss $\ell_i^{\mathrm{dev}}(a,r)$ and the bounded trajectory diagnostics required by the protocol. The main comparison uses participating laboratories' panels; fixed-pool repartitioning retains all ten original panels; source-transfer studies use the original target panels. The coordinator computes

$$\bar{s}(a,r) = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} s_i(a,r), \qquad \bar{\ell}(a,r) = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \ell_i^{\mathrm{dev}}(a,r). \tag{2}$$

The selected round $\widehat{r}_a$ maximizes $\bar{s}(a,r)$, with lower $\bar{\ell}(a,r)$ breaking ties and the earlier round retained on a complete tie. Its parameters are $\widehat{w}_a = w_{\widehat{r}_a}$. The pair $q(a) = (\bar{s}(a,\widehat{r}_a), -\bar{\ell}(a,\widehat{r}_a))$ ranks the design. We call $\bar{s}(a,\widehat{r}_a)$ the *dev score*, labeled "highest dev score" in Figure 2. Training weights examples, while original-protocol selection weights the designated panels equally. The original searches use 100 rounds per candidate; Appendices E.6 and E.9 specify the follow-up batch sizes, 20-round schedules and development scorers, including pooled AUROC/AP in E.9. Each trial produces a *development-evidence card*. After t trials, the sequence $\mathcal{E}_t = (e_1, \ldots, e_t)$ forms the accumulated *development evidence*. For $1 \le j \le t$, card e_j records the executed configuration a_j, hypothesis h_j, best development primary score and loss (s_j, ℓ_j), selected round $\widehat{r}_j$ and acceptance decision dec_j. Endpoint diagnostics Δ_j contain the first and last development-evaluation rounds, their sample-weighted training losses, equal-laboratory development scores and losses, and the changes between those measurements, plus the optional worst-laboratory score and evaluating-laboratory count.

For auxiliary-source studies with different prediction targets, shared encoders follow the same update rule, while each source-specific head is aggregated only over laboratories using that prediction target. Checkpoint selection uses the original target's development panels. Each source retains its own prediction objective while contributing updates to the shared representation.

3.3 Proposal revision from development evidence

The first proposal is generated once without development feedback and shared by direct and loop search. History feedback begins with the second proposal. For subsequent research steps t, let a_t be the retained design, $\mathcal{E}_t$ the development evidence accumulated so far and $\mathcal{U}_t$ the untried configurations. The research model S, with weights θ held fixed, generates a proposal p_t and the coordinator maps its `design_id`, denoted $\mathrm{id}(p_t)$, to the candidate:

$$p_t \sim P_\theta(\cdot \mid \mathcal{T}, a_t, \mathcal{E}_t, \mathcal{U}_t), \qquad \widetilde{a}_t = \mathrm{Compile}(\mathrm{id}(p_t)). \tag{3}$$

The proposal also contains a `hypothesis`, an `experiment` and an `expected_effect`, recording the motivation, intended comparison and anticipated outcome. The identifier determines execution; measured development performance determines selection.

After training $\widetilde{a}_t$, the coordinator constructs the next card e_{t+1}. Comparing q lexicographically by higher score and then lower loss gives

$$a_{t+1} = \begin{cases} \widetilde{a}_t, & q(\widetilde{a}_t) >_{\mathrm{lex}} q(a_t), \\ a_t, & \text{otherwise}, \end{cases} \qquad \mathcal{E}_{t+1} = \mathcal{E}_t \parallel [e_{t+1}], \tag{4}$$

where $\parallel$ appends a card to the history. Unsuccessful trials remain in the history. Invalid proposals and numerically failed fits consume a proposal slot and leave the incumbent unchanged. Updated context carries these outcomes into subsequent proposals. After the proposal budget is exhausted, the retained design and its selected checkpoint are evaluated on the held-out task.

3.4 Evidence-guided training allocation

We use trajectory-guided allocation in a controlled study (Section 4.4) to test whether early evaluation evidence improves the use of a fixed training budget. The screening–promotion policy retains both high-scoring and still-improving candidates. It screens ten predetermined designs for 20 rounds, evaluating every five rounds through Equation 2. Within each of three learning-rate groups, it promotes the best-scoring design and, among the remaining designs, the one with the largest score improvement from round 5 to round 20. Development loss breaks score ties. The six promoted designs restart from their initial parameters for 100-round training, and their best development checkpoints determine the final predictor.

The two promotion criteria reserve training capacity for candidates at different stages of convergence. Current score favors a strong screening checkpoint, while improvement over the shared screening interval retains a candidate whose score is still rising. Grouping by learning rate preserves representation from each group in the full-training stage. All promotion decisions use development measurements.

The control trains all ten designs for 80 rounds. Both schedules use 800 aggregation rounds and 160 aggregate development evaluations. This study evaluates allocation over a predetermined candidate set independently of proposal generation. The main comparison and proposal-history experiments use 100 training rounds per candidate. Section 4.4 reports the allocation results; Appendix A provides the complete proposal schema, algorithm and allocation rules.

4. Experiments

4.1 Tasks, datasets and evaluation

We evaluate BioCoLoop on three biological prediction problems with task-specific base models and a common procedure for collaborative training, evaluation and proposal generation. The main setting partitions each dataset into ten disjoint groups. Complementary studies use distinct biological datasets or experimental scenarios as laboratories to evaluate source transfer.

Drug–target interactions. Given a compound and a protein sequence, the model predicts whether they interact. BindingDB provides 50,149 training pairs and 5,604 development pairs, with canonical compounds defining laboratory assignments. We use TAPB as the base model, with frozen ESM2 protein features and trainable drug, fusion and classification modules (Lin et al., 2025). The primary metric is AUROC on 5,505 random test pairs. Additional test sets containing 3,791 unseen-drug pairs and 2,507 unseen-protein pairs measure generalization to unseen entities. AP, MCC, accuracy and cross-entropy are reported as secondary metrics.

Observed-response proteomics. Given a drug representation and its measured 6h and 24h proteomic responses, the model predicts binary efficacy. A ProteinTalks PTPC subset provides 386 training, 105 development and 148 test conditions (Sun et al., 2026). Compound-group assignment gives each laboratory 49-50 training and development conditions. The base model is a ProteinTalks-derived efficacy model with protein and drug branches, trained from random initialization. Average precision (AP) is the primary metric. AUROC, accuracy and cross-entropy are secondary metrics. Unlike de novo response prediction, this task conditions on observed post-treatment proteomic measurements.

Cell perturbation identification. Given an expression response and five candidate interventions, the model identifies the intervention whose predicted response has the highest cosine similarity to the observation. We use VCC single-gene perturbations (Arc Institute, 2025), Norman double-gene perturbations (Norman et al., 2019) and a Tahoe subset of drug perturbations in A549 at 5 μM (Zhang et al., 2025). Intervention-disjoint training, development and test counts are 60/20/20, 70/20/27 and 180/60/67, respectively. The base model uses a frozen scDEBART backbone with a trainable response head (Sung & Kim, 2026). All methods use the same corrected attention masks and the same candidate sets. The primary metric is intervention-macro Top-1, which first averages accuracy within each intervention and then across interventions. Random selection has an expected Top-1 of 20%. Micro Top-1, MRR, Top-3 and cross-entropy are reported as secondary metrics. Appendix A provides the response construction, chemical conditioning, training losses, and secondary evaluation details.

4.2 Compared methods and shared protocol

We compare BioCoLoop with three types of baselines. The *fixed model* fits the task-specific base model once with a predefined configuration. *Direct optimization* uses BioCoLoop's proposal generator and registered executable design space, generating proposals without previous development outcomes. AI-Scientist-v2 (Yamada et al., 2025) and AI-Researcher (Tang et al., 2025) are adapted to the same biological tasks and fitting interface. Every method returns one predictor; laboratories contribute to its fitting, not an inference-time ensemble. *Direct* means feedback-free proposal generation, not a single training run; the fixed model is the single-fit reference.

BioCoLoop combines collaborative fitting and evaluation with development-history feedback. We use Qwen2.5-7B-Instruct and GPT-5.6 Luna as proposal generators (OpenAI, 2026). **Evaluation protocol.** We simulate data held by separate laboratories by partitioning each training pool into disjoint laboratory shards, as in synthetic federated partitions (McMahan et al., 2017; He et al., 2020b); benchmarks such as LEAF instead use natural client partitions (Caldas et al., 2018), and our source-defined study (Section 4.5) follows that alternative. As in federated studies in medicine and biology (Sheller et al., 2020; Chang et al., 2018; Dayan et al., 2021; Ogier du Terrail et al., 2022; Warnat-Herresthal et al., 2021; Wang et al., 2024), the contribution of collaboration is measured by comparing ten collaborating laboratories with laboratory 0 developing models alone; all methods are scored on the same task-specific held-out test, which never enters training, selection or proposal feedback. Section 4.7 and Appendix E additionally fix laboratories, panels, design choices and candidate budgets to characterize search at matched access.

The main comparison uses twelve configurations and at most six proposals beyond the initial design. Each valid candidate starts from the prescribed task initialization and receives 100 training rounds. Development score and loss determine the retained design and checkpoint before held-out scoring. We report means and sample standard deviations over seeds 42–44 on fixed data partitions. These summaries describe training and search variability; paired cluster-bootstrap intervals in the Statistical Supplement quantify held-out sample uncertainty. The allocation, source-transfer and extended-search studies use their separately specified budgets and seeds.

4.3 Collaborative versus single-laboratory development

Collaboration produces substantially stronger models. Running the same history-enabled Qwen workflow with laboratory 0 alone and with ten laboratories, collaboration improves every primary endpoint in mean on the common held-out tests: DTI AUROC 89.89→93.76, proteomic AP 20.13→36.99, VCC Top-1 20.22→31.54, Norman 14.15→22.25 and Tahoe 15.42→22.39 (Appendix Table D.1). The gain is not recovered by searching harder alone: single-laboratory random, Bayesian and exhaustive twelve-design search remain below the collaborative result in mean on all five endpoints, and collaboration also improves DTI on unseen drugs and targets (Appendix D).

A fixed default design trained with all ten laboratories separates shared data from collaborative design search (Appendix Table D.1). On DTI, proteomics and VCC, shared data account for most of the gain: the fixed model rises from 82.83 to 91.80 on DTI, 25.05 to 37.07 on proteomics and 29.94 to 31.54 on VCC. Collaborative search then adds 1.96 AUROC on DTI and 6.77 Top-1 points on Norman, where the default design falls from 25.56 with one laboratory to 15.48 with ten; it nearly matches the fixed model on proteomics (36.99 versus 37.07 AP), ties it on VCC and is 3.48 points lower on Tahoe, where a development-selected design transfers less well (Appendix E.3 analyzes this divergence).

Table 1 follows the federated convention of comparing collaborative with local-only development, and places BioCoLoop with ten laboratories beside the single-laboratory fixed model, direct search and two public research agents. Its reference row is therefore the single-laboratory fixed model rather than the single-laboratory BioCoLoop run used above. With Qwen, BioCoLoop reaches 93.76 AUROC on BindingDB and 36.99 AP on proteomics, compared with fixed-model scores of 82.83 and 25.05; Luna reaches 93.67 and 37.11. BioCoLoop also leads the VCC and Tahoe comparisons in both blocks. On Norman, the single-laboratory fixed configuration scores 25.56 versus 22.25/22.69; Appendix E.7 reports fixed-model variation across laboratory identities. The interface also accepts other controllers: in a seed-42 pilot with ten laboratories, AI-Scientist-v2 matches BioCoLoop on proteomics, VCC, Norman and Tahoe and reaches 95.07 versus 94.60 on DTI (Appendix C).

Table 1: Collaboration-versus-local reference: BioCoLoop with ten participating laboratories beside workflows run by one laboratory, the usual federated comparison of collaborative and local-only development. Scores $\times 100$ are mean $\pm$ sample SD over seeds 42–44. Data and local computation differ with access, so rows across the two access levels compare complete scenarios rather than search methods; Appendix Table D.1 adds a ten-laboratory fixed model. Cell Top-1 chooses among five candidates (chance: 20%). Bold and underline indicate the best and second-best complete scores within each block. †: two scored seeds; –: no sealed score; ⋆: extended-context adaptation. Execution outcomes are in Appendix C; same-access search is in Section 4.7.

Method	Labs	DTI	Proteomics	Cell perturbation (Top-1)		
		AUROC ↑	AP ↑	VCC ↑	Norman ↑	Tahoe ↑
Qwen2.5-7B-Instruct						
Fixed model	1	82.83 ± 1.22	<u>25.05</u> ± 4.56	<u>29.94</u> ± 0.00	**25.56** ± 0.00	<u>19.40</u> ± 0.00
Direct	1	<u>89.89</u> ± 1.02	20.13 ± 0.07	21.09 ± 4.38	14.15 ± 0.00	15.42 ± 6.21
AI-Scientist-v2	1	89.68 ± 0.80	20.18 ± 0.43	18.61 ± 2.78	14.12 ± 0.04	15.42 ± 6.21
AI-Researcher	1	86.09 ± 0.88	20.12 ± 0.05	19.70 ± 2.87†	21.73 ± 6.63	17.41 ± 6.21
BioCoLoop	10	**93.76** ± 1.11	**36.99** ± 0.22	**31.54** ± 0.08	<u>22.25</u> ± 0.45	**22.39** ± 2.59
GPT-5.6 Luna (low reasoning)						
Fixed model	1	82.83 ± 1.22	<u>25.05</u> ± 4.56	<u>29.94</u> ± 0.00	**25.56** ± 0.00	<u>19.40</u> ± 0.00
Direct	1	<u>89.89</u> ± 1.02	20.17 ± 0.42	18.61 ± 2.78	21.75 ± 6.59	15.92 ± 6.03
AI-Scientist-v2	1	89.59 ± 0.95	23.37 ± 5.89	18.61 ± 2.78	14.12 ± 0.04	11.94 ± 2.11†
AI-Researcher	1	–	20.29 ± 0.27⋆	18.61 ± 2.78⋆	14.12 ± 0.04⋆	15.42 ± 6.21⋆
BioCoLoop	10	**93.67** ± 0.98	**37.11** ± 0.01	**31.50** ± 0.00	<u>22.69</u> ± 0.73	**22.39** ± 2.59

Table 2: Evidence-guided training allocation at a matched budget. Held-out scores use the same ten designs, ten laboratories, 800 full-client aggregation rounds and 160 aggregate development evaluations per method. Uniform allocation gives every design 80 rounds; the evidence-guided scheduler screens all designs for 20 rounds and restarts six promoted designs for 100-round training. Gain is evidence-guided minus uniform, in percentage points; W/T/L counts seed-level wins, ties and losses (total 4/8/0). Norman 95% CI: [4.54, 17.73]; Tahoe: [−1.00, 7.46]. The separate DTI development replay is in Appendix A.

Benchmark	**Uniform**	**Evidence-guided**	**Gain**	**W/T/L**
Proteomics: efficacy	**37.47**	**37.47**	+0.00	0/3/0
Cell: VCC single-gene	**20.01**	**20.01**	+0.00	0/3/0
Cell: Norman double-gene	11.48	**22.32**	**+10.84**	3/0/0
Cell: Tahoe drug	20.90	**22.89**	**+1.99**	1/2/0

4.4 Evidence-guided training allocation

Evidence-guided allocation improves held-out performance at a matched budget (Table 2). Uniform allocation trains all ten designs for 80 rounds. The evidence-guided policy screens them for 20 rounds and restarts six selected designs for 100-round training. Each schedule uses 800 aggregation rounds and 160 development evaluations.

Norman Top-1 increases from 11.48% to 22.32%, a 10.84-point gain with a hierarchical paired-bootstrap 95% interval of [4.54, 17.73]. A selected low-learning-rate model improves beyond the uniform stopping point. Across the twelve task–seed pairs, allocation yields four wins, eight ties and no losses: three gains on Norman and one on Tahoe. Screening and restarting costs are included. This fixed-slate experiment isolates training allocation from language-model proposal history.

A fresh 360-fit study adds score-only and random promotion (Table E.5). On Tahoe, trajectory promotion reaches 22.89 versus 20.90 for score-only and 21.95 for random. On Norman, score-only reaches 21.70 and random 22.63, versus our 22.32; random also leads on VCC (22.49 versus 20.01). Against random promotion, trajectory promotion wins six and loses six of the twelve task–seed pairs (mean -0.44 points), while it wins four pairs and loses none against uniform allocation, and wins four and loses one against score-only promotion. Screening followed by selective longer fitting

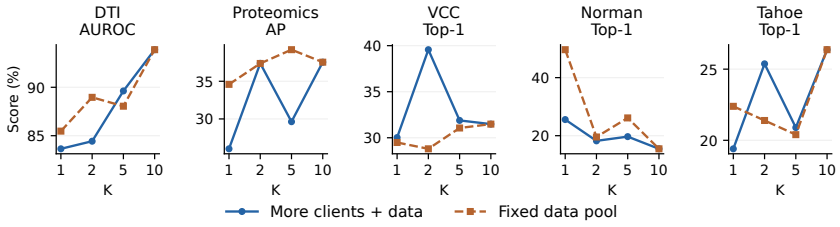


Figure 3: Effect of the number of laboratories. Solid lines add both laboratories and training data. Dashed lines divide the same training data across different numbers of laboratories. Full results are reported in Appendix B.

therefore explains most of the gain over uniform allocation; the trajectory rule adds robustness over score-only promotion, and its choice among promotion rules has task-dependent effects.

4.5 Learning from different data sources

We also test collaboration when laboratories hold different biological studies rather than shards of one pool. We combine BindingDB with BioSNAP, Davis and Human; VCC with three external single-cell studies; and PTPC with Lin, Ruprecht and decryptE (Lin et al., 2026; Ruprecht et al., 2020; Eckert et al., 2025). Target development data select each model, and all configurations use the same target test set.

With one proteomic study per laboratory, the three auxiliary sources raise target AUROC from 72.06 to 74.13 over eight seeds (paired $+2.07$ points, 95% seed-block interval [0.65, 3.38], higher in seven of eight seeds) and AP from 34.60 to 36.82 ($+2.22$, $[-1.19, 5.59]$, higher in four of eight). AP improves in seeds 61–64 and decreases in seeds 65–68 (-5.87 to -0.23), so the AP benefit is seed-dependent; the gain comes mainly from decryptE, and Lin and Ruprecht alone leave the target near its own level. Shared encoders combine local updates while source-specific heads retain their own objectives. Benefits follow source compatibility. Davis and Human raise target DTI AUROC from 85.47 to 87.86 and 87.95, whereas BioSNAP lowers it by 3.03 and all three sources together by 1.18 points; Jiang raises VCC Top-1 from 29.49 to 33.07, whereas Replogle, Nadig and all three external studies lower it by 2.15–2.55 points. At four laboratories and 60 training interventions in both cell arms, same-source VCC reaches 30.70% Top-1 versus 28.82% with external studies. Appendix B reports every source comparison; Appendix E.4 tests proposal history at matched source access.

4.6 Effect of the number of laboratories

We vary $K \in \{1, 2, 5, 10\}$ by adding laboratories and data or repartitioning a fixed pool (Figure 3). With the default design, adding participating laboratories, and with them data, raises DTI, proteomics, VCC and Tahoe above the single-laboratory score, with VCC peaking at $K = 2$; Norman instead falls from 25.56 to 15.48 Top-1, the case where collaborative design search recovers 6.77 points (Section 4.3). When a fixed pool is merely repartitioned, effects depend on the task: proteomics peaks at an intermediate K and Norman degrades as its pool is split further.

Both settings use the default design and training schedule. Increasing participation adds local development panels; fixed-pool repartitioning keeps all ten original evaluation panels. In cell tasks, training groups also define contrastive-negative pools. Appendix B.1 reports these optimization and evaluation differences alongside the complete results.

4.7 SAME-ACCESS CONFIGURATION SEARCH

Table 3 compares separate studies at matched access and candidate budgets, each over eight training/controller runs (three original seeds and five registered extensions): 576 expanded-recipe DTI/protein fits (Appendix E.9) and 128 gene-disjoint cell fits (E.6). Each uses its own fixed protocol and complements the original single-seed matched-access study (Appendix B.3), in which history and direct proposals tie on eight of ten endpoint–proposer pairs, with one gain (Norman/Luna, $+0.81$ Top-1) and one loss (DTI/Luna, -0.33 AUROC). In the separate 36-design, 24-slot study, final history and direct scores tie on seven of eight comparisons; history is lower on VCC/Luna (20.01 versus 20.23).

Table 3: Matched-access search in three separate protocols (eight training/controller runs: the original three plus five registered extensions; mean $\pm$ sample SD, scores $\times 100$). DTI/protein: Luna, 36 recipes, ten laboratories, one historical starting predictor per task. Cell: Qwen, 16 designs, four study–cell contexts and gene-disjoint partitions. Searches allow six requests after D00 and 20 rounds per valid fit; comparisons are within columns. Complete feedback, proposer and GP controls appear in E.6/E.9. Bold marks the best means, including ties.

Policy	DTI AUROC	Protein AP	Cell Top-1
Fitted D00	94.271 ± 0.000	37.109 ± 0.000	21.370 ± 0.217
Direct	$\mathbf{94.372 \pm 0.005}$	38.994 ± 0.537	22.344 ± 0.506
Structured history	$\mathbf{94.372 \pm 0.005}$	38.789 ± 0.644	22.753 ± 0.528
Random expectation	94.319 ± 0.006	38.581 ± 0.381	22.692 ± 0.206
GP–EI	94.286 ± 0.021	$\mathbf{39.016 \pm 0.544}$	$\mathbf{23.278 \pm 0.344}$

On DTI, Luna structured history reaches 94.372 AUROC versus 94.319 for random, a paired difference of $+0.053$ points (seed-block 95% interval [0.050, 0.055]; all eight runs improve), and exceeds GP–EI in every run. On protein, direct search (38.994 AP) and GP–EI (39.016) have the highest means; structured history (38.789) is above random (38.581) in six of eight runs ($+0.21$, $[-0.04, 0.44]$). On the cell benchmark, structured history reaches 22.753 Top-1, above direct search (22.344; $+0.41$, [0.09, 0.84]) and close to random (22.692; $+0.06$, $[-0.33, 0.46]$), while GP–EI is highest (23.278). Language-model-guided variants thus exceed the random expectation in mean in all three task families, by small margins that vary by task; random scores are exact single-predictor expectations. Proposal history is task-dependent: on DTI, Luna's direct and history modes select the same design in every run, so the DTI gain over random comes from language-model proposals rather than history; on cells, history improves on feedback-free proposals; on protein, feedback-free and Bayesian search are at least as strong. Longer searches can also improve development scores while reducing held-out performance (Qwen VCC/Tahoe: declines of 11.89/11.94 points); the audits in E.3/E.8 show selection sensitivity without isolating its cause. With source-defined laboratories (Appendix E.4), no search policy exceeds the fixed design and Qwen history is below direct search.

5. CONCLUSION

BioCoLoop makes biological model development collaborative without pooling raw data. Compared with a laboratory working alone on common held-out tests, ten-laboratory development improves all five primary endpoints in mean, beyond single-laboratory conventional search; distinct proteomic studies improve target ranking, and evidence-guided allocation improves perturbation identification at a matched budget. Collaboration is simulated on shared compute; stabilizing development selection and real multi-site deployment are natural next steps.

REPRODUCIBILITY STATEMENT

Appendix A specifies task construction, optimization, proposal schemas and scorers. Appendices B–D document source, sensitivity, public-agent and single-laboratory comparisons; Appendix E gives matched search, allocation and validation protocols. Records retain data/code hashes, seeds, proposals, development trajectories, selection seals and saved predictions. Independent checks reproduce the scored models. The original searches retain all 504 attempted slots; the follow-ups

include 360 allocation fits, 128 gene-disjoint cell fits over eight seeds, 40 laboratory-identity fits, the 72-checkpoint panel audit and 216 expanded-recipe fits with 360 more in their eight-seed extension. Pilot and interrupted engineering runs are recorded separately. The Statistical Supplement retains complete original per-seed contrasts, including zero and negative differences. Random-policy variability, training/controller variation and model-conditional test-sample intervals are reported separately.

ETHICS STATEMENT

This computational study uses previously released biological resources under their source access conditions and collects no new measurements or participant data. Raw data remain local in the simulated laboratory protocol, which provides no formal privacy guarantee. Multi-institutional deployment would require a privacy and security policy for parameter updates and aggregate diagnostics. Predictions are evaluated for research use and have no biological or therapeutic validation.

AI USE STATEMENT

The main comparison retains matched proposal records for local Qwen2.5-7B-Instruct and the service identified as `gpt-5.6-luna`. Luna is run at low reasoning effort through a text-only bridge with tools disabled. Its service does not expose a generation seed, so the seed labels identify the training/search repetitions rather than deterministic language-model sampling. AI assistants contributed to hypothesis refinement, experimental protocol and software development, data processing and checks, result analysis and interpretation, literature retrieval, figures, and manuscript drafting and revision. The research artifacts retain the generated proposals and their measured outcomes. The authors are responsible for verifying the final methods, results, citations and manuscript, including AI-assisted material.

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APPENDIX A. ALGORITHM AND TASK PROTOCOLS

How to read the supplementary experiments. Appendix A specifies the algorithm and scorers; B examines laboratory count, source transfer and proposal history; C documents task-adapted public agents; D compares single-laboratory search; and E compares same-access search and training promotion and diagnoses development–test divergence. Ten-versus-one laboratory comparisons measure the complete access scenario, whereas same-access controls isolate research decisions.

Study	What changes	What the comparison measures
Access scenario	One versus ten laboratories and research workflow	Combined data-access and workflow effect
Same-access search	Proposal policy; common data and menu	Configuration-search quality
Source transfer	Biological study supplying local updates	Positive or negative transfer to the target
Training allocation	Which candidates receive longer fitting	Use of a fixed training/evaluation budget
Roster diagnostic	Development distractors and option order	Sensitivity of already selected models

A.1 RESEARCH ALGORITHM AND OPTIMIZATION

All task families share Section 3's parameter update, development selection and proposal interface. Adapters define task-specific representations, prediction components, losses and scorers. The fitted reference design supplies the first evidence card.

	BioCoLoop research procedure
1	Fit the reference from the task initialization and seed; retain its best development checkpoint and initialize the history.
2	Generate the shared first proposal without development feedback. Subsequent proposals receive the task, untried configurations, retained design and evidence cards.
3	Map `design_id` to its executable configuration, allowing at most two generation attempts per slot.
4	Fit each valid candidate from the same initialization procedure and seed for its assigned aggregation rounds.
5	Select its best development checkpoint; replace the incumbent only for a better score/loss pair.
6	Append the executed configuration, hypothesis, diagnostics and decision, including rejected or failed trials.
7	Return the final retained design, checkpoint and history; evaluate that checkpoint on the held-out task.

Direct and history-guided search share the first feedback-free proposal. Thereafter, direct receives the default configuration and untried identifiers without outcomes; history-guided search receives the retained configuration and accumulated cards. Both share the registered design space, proposal count and training schedule; Appendix B reports every prespecified prefix. A completed configuration fit may be reused only with identical task, data access and seed, retaining its logical position and budget charge.

Each aggregation round runs one local AdamW epoch per participating laboratory, with batch size 64, gradient-norm clipping at one and optimizer-state reset. Designs specify learning rate and weight decay; the proximal penalty is always zero. Parameter aggregation is training-example-weighted, with coordinator momentum applied only to trainable parameters (Equation 1). Floating-point buffers use the same weights; integer and Boolean buffers use elementwise maxima. Candidates share the task-specific public or random initialization procedure and seed. The optional width-32 residual module has zero-initialized output; the original head remains trainable. Code uses *client*/*worker* for laboratory and `server_momentum` for coordinator momentum.

Full fits use 100 rounds and development evaluation every five rounds. Selection maximizes mean primary score, then minimizes mean loss over the designated panels (Section 3.2): the participating laboratories in the main experiment. Complete ties retain the earlier checkpoint or incumbent. Training counts refer to adapter rows—intervention-level profiles, not underlying cell counts, for cell tasks.

A.2 ORIGINAL EXECUTABLE DESIGN SPACE, PROPOSALS AND EVIDENCE

Each `D` identifier specifies a four-factor configuration below. Starting from D00, the main study selects six distinct designs from D01–D11. Appendix B defines the extended design space.

Design	Learning rate	Weight decay	Momentum	Residual module
D00	0.001	0.001	0	no
D01	0.0001	0.001	0	no
D02	0.01	0.001	0	no
D03	0.001	0	0	no
D04	0.001	0.01	0	no
D05	0.001	0.001	0.9	no
D06	0.001	0.001	0	yes
D07	0.001	0.001	0.9	yes
D08	0.0001	0.001	0	yes
D09	0.0001	0.001	0.9	yes
D10	0.01	0.01	0	no
D11	0.001	0.01	0	yes

Qwen2.5-7B-Instruct uses bfloat16 inference, temperature 0.7, top-p 0.9 and at most 512 new tokens. Its sampling seed is $z + t + 10{,}000b$ for study seed z, zero-indexed slot t and attempt $b \in \{0, 1\}$. GPT-5.6 Luna uses a text-only bridge at low reasoning effort, tools disabled and the same proposal protocol. It exposes no generation seed; labels identify training/search repetitions. Invalid identifiers, repeats or malformed responses trigger a second attempt; another failure consumes the slot without a fit. Numerically failed fits retain the incumbent and remain in history.

A proposal has exactly four fields:

```
{
  "hypothesis": "why the proposed change may help",
  "experiment": "the intended controlled comparison",
  "expected_effect": "expected metric or diagnostic change",
  "design_id": "D08"
}
```

The descriptive fields are nonempty rationale strings; the registered identifier determines the executed configuration.

Evidence cards contain the configuration, hypothesis, best development score/loss, selected round and acceptance decision. Trajectory summaries record the first and last evaluation rounds (5 and 100 for full fits), their sample-weighted training losses, equal-laboratory development scores/losses and last-minus-first changes in both development metrics. Main-study summaries also include the worst laboratory score and evaluating-laboratory count. Source-transfer and repartitioning studies specify training participation and evaluation panels separately. Cards remain ordered, including unsuccessful trials.

An observed proposal sequence. For Norman with ten laboratories and seed 42, first-slot D06 ties D00's development Top-1 of 40.00 but lowers loss from 158.98 to 158.94 and is accepted. Second-slot D08 retains the residual module and reduces learning rate from 10^{-3} to 10^{-4}. Between rounds 5 and 100, its Top-1 rises from 10.20 to 51.90 and loss falls from 168.12 to 153.85. It is accepted; the next design, D07, reaches only 45.00 and is rejected. Both outcomes remain in history.

A.3 Independent training-allocation control

This study varies training-round allocation over ten fixed configurations and ten laboratories, using seeds 53–55 on proteomics and the three cell settings.

Learning rate	Weight decay	Momentum	Residual module
0.001	0.001	0	no
0.0001	0.001	0	no
0.0001	0.001	0	yes
0.0001	0.01	0.9	no
0.0001	0.01	0.9	yes
0.001	0.001	0.9	yes
0.01	0.001	0	yes
0.01	0.001	0.9	yes
0.01	0.01	0.9	no
0.001	0.01	0	yes

Uniform allocation fits each design for 80 rounds. Trajectory-guided allocation screens each for 20, then promotes two designs from each of three learning-rate groups: the design with the best screening-checkpoint primary score (lower loss breaks ties), and the remaining design with the largest round-5-to-20 score improvement (lower round-20 loss breaks ties). Complete ties follow table order. The six promoted designs restart from initialization for 100 rounds; final selection uses only their full-training checkpoints. The other four stop after screening.

Both schedules use 800 aggregation rounds, $10 \times 80 = 10 \times 20 + 6 \times 100$, or 8,000 local epochs. Evaluation every five rounds gives 160 aggregate evaluations of ten laboratories each. Training/evaluation counts are matched; the scheduler generates no language-model proposals.

Section 4.4 reports the paired held-out outcomes; Appendix E.5 gives the complete promotion controls. A separate TAPB development replay has unequal candidate-round budgets and is excluded from the held-out allocation evidence.

A.4 Data construction and task adapters

Drug–target interaction. BindingDB contains 50,149 fitting and 5,604 development pairs. Assigning canonical compounds to ten laboratories gives 4,932–5,058 training and 517–643 development pairs each. Test sets contain 5,505 random, 3,791 unseen-drug and 2,507 unseen-protein pairs. TAPB combines frozen public protein features with trainable drug, fusion and classification components, optionally adding a residual adapter before confounder-averaged prediction. Its objective combines classification NLL and masked-language loss.

Observed-response proteomics. PTPC has 333 compound groups and 491 conditions: 386 fitting (35–42 per laboratory) and 105 development conditions; the test set has 148. Each 6h/24h profile undergoes log1p, within-profile min–max normalization over finite observations, zero filling and unit L2 normalization. The 2,048-bit Morgan vector is L2-normalized and scaled by 0.5. Two 5,171-dimensional protein blocks plus drug features form a 12,390-dimensional float32 input. The ProteinTalks-derived head combines protein/drug convolutional branches and a 64–32–1 classifier (464,193 D00 parameters), trained from random initialization with binary cross-entropy. All methods share these inputs and architecture; selection uses AP.

Cell inputs and response units. VCC, Norman and Tahoe use whole-intervention fitting/development/test splits of 60/20/20, 70/20/27 and 180/60/67. Training uses one mean profile per intervention. VCC development/test queries average disjoint groups of up to 64 cells, omitting a final group smaller than 16. Norman uses released scVI-derived differential-expression profiles per gene pair; Tahoe uses one aggregate profile per compound at the selected cell context and dose.

scDEBART encodes the shared untreated reference and candidate perturbation; its features are frozen while the response head trains. The corrected extractor implements three additive blocking masks, preserving pretrained weights, gene panels, targets, partitions and candidate options. Genetic scoring excludes the union of directly perturbed genes. Tahoe appends a fixed 128-dimensional projection (seed 1234) of a 2,048-bit Morgan fingerprint to 768-dimensional backbone features, zero-initializing the additional input weights. Training minimizes response MSE plus 0.1 times local-minibatch intervention contrastive loss, with cosine-logit temperature 0.1.

Five-option queries and scoring. Each development/test intervention receives four distinct distractors sampled uniformly without replacement from its *own split*, excluding the true intervention. After adding the true option, positions are shuffled. This roster is fixed across its response queries, training seeds and methods. Split and roster generation use seed 20260918. Training instead draws contrastive negatives from the local minibatch, not these rosters.

Candidates are ranked by cosine similarity between predicted and observed responses, L2-normalized over scoring genes; ties retain shuffled option order. For G interventions, m_g queries per intervention g, and true-option rank $r_{g,j}$ in query j, the primary score is

$$\mathrm{MacroTop1} = \frac{1}{G}\sum_{g=1}^{G}\frac{1}{m_g}\sum_{j=1}^{m_g}\mathbf{1}\{r_{g,j}=1\}.$$

Interventions receive equal weight regardless of query count. Micro Top-1, mean reciprocal rank and Top-3 average over queries. Development loss is cross-entropy over the five cosine scores at temperature 0.1. Selection computes metrics within each laboratory, then averages laboratories equally.

Identification diagnostics. Frozen-scDEBART diagnostics span 7.5–31.6% macro Top-1 (chance: 20%) and 36.00–57.00 MRR (chance: 45.67%, from $(1+1/2+1/3+1/4+1/5)/5$). The untrained public head scores 7.5% on Norman. True-option positions are near-uniform; an oracle constructed from observed held-out response prototypes retrieves 94.5% on VCC and 100% on Norman.

A.5 Complete comparisons and uncertainty

The tables report controls, secondary metrics and all three DTI test sets. Bold marks the best task/metric values, including ties; higher is better except for binary and cell cross-entropy.

Three-seed means and sample SDs describe training/search variability on fixed partitions. Sample-level intervals use 1,000 fixed-seed paired cluster resamples of retained predictions: drug identity for random/unseen-drug DTI, protein sequence for unseen-protein DTI, compounds for proteomics, and interventions for cells. Contrasted methods share resampled clusters. Allocation uses 5,000 hierarchical paired resamples, first drawing training seeds and then compound/intervention groups within each sampled seed.

BioCoLoop Statistical Supplement, Tables S.1–S.2, reports all per-seed differences and intervals for the main contrasts and three DTI test sets, including zero and negative differences. These model-conditional intervals describe test-sample uncertainty; several cell intervals include zero.

Interpreting the single-laboratory control. The collaboration effect is defined as in the federated evaluations cited in Section 4.2: the same workflow with all participating laboratories versus a laboratory restricted to its own data, both scored on one held-out test that neither uses for selection. Laboratory 0 is the pre-specified local reference; Appendix E.7 reports fixed-model fits for every laboratory identity, which vary with shard composition (e.g., VCC). Appendix D.1 explains why single-laboratory search, including BioCoLoop, often selects designs whose local-development ranking does not transfer to the held-out set.

Configurations, proposals, development histories and saved predictions support reconstruction; the reproducibility statement details their availability and verification.

Table A.2: Held-out secondary metrics for the task-reference models, averaged over seeds 42–44 and multiplied by 100. Bold indicates all tied maxima, or minima for loss.

Method	Labs	Metric 1	Metric 2	Metric 3	Metric 4
DTI ($n = 3$)		AUROC	AP	MCC	BCE
Fixed model	1	82.83	81.57	49.95	52.37
Direct	1	89.89	87.77	65.71	61.02
Our loop	1	89.89	87.77	65.71	61.02
Fixed model	10	91.80	89.36	72.39	**36.54**

Method	Labs	Metric 1	Metric 2	Metric 3	Metric 4
Direct	10	**93.76**	**92.73**	**73.71**	36.59
Our loop	10	**93.76**	**92.73**	**73.71**	36.59
Proteomics ($n = 3$)		AP	AUROC	BCE	Acc.
Fixed model	1	25.05	65.49	42.65	**85.14**
Direct	1	20.13	61.80	44.34	81.76
Our loop	1	20.13	61.80	44.34	81.76
Fixed model	10	**37.07**	**74.06**	**39.78**	**85.14**
Direct	10	36.99	73.87	39.83	**85.14**
Our loop	10	36.99	73.87	39.83	**85.14**
VCC ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	33.61	**56.53**	**73.89**	**154.51**
Direct	1	21.48	47.10	62.78	160.82
Our loop	1	20.74	46.94	64.26	160.86
Fixed model	10	**33.98**	56.34	69.44	158.14
Direct	10	**33.98**	56.34	69.44	158.14
Our loop	10	**33.98**	56.34	69.44	158.14
Norman ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	**25.56**	48.50	63.93	158.74
Direct	1	14.15	42.12	58.35	160.39
Our loop	1	14.15	42.12	58.35	160.39
Fixed model	10	15.48	45.26	65.63	**158.06**
Direct	10	22.25	**50.07**	**70.96**	158.83
Our loop	10	22.25	**50.07**	**70.96**	158.83
Tahoe ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	19.40	45.00	58.21	163.08
Direct	1	15.42	42.60	56.72	161.90
Our loop	1	15.42	42.60	56.72	161.90
Fixed model	10	**25.87**	**50.20**	**65.17**	**160.61**
Direct	10	22.39	47.99	61.69	160.75
Our loop	10	22.39	47.95	61.19	160.75

Table A.3: All three frozen TAPB held-out endpoints under the committed common protocol. Scores are completed-seed means multiplied by 100; N/A means no complete independently verified evaluation. The main DTI endpoint is random held-out.

Endpoint	Method	Labs	AUROC	AP
Random	Fixed model	1	82.83	81.57
Random	Direct	1	89.89	87.77
Random	Our loop	1	89.89	87.77
Random	Fixed model	10	91.80	89.36
Random	Direct	10	**93.76**	**92.73**
Random	Our loop	10	**93.76**	**92.73**
Unseen drug	Fixed model	1	81.96	85.78
Unseen drug	Direct	1	88.39	89.35
Unseen drug	Our loop	1	88.39	89.35
Unseen drug	Fixed model	10	90.06	90.40
Unseen drug	Direct	10	**91.65**	**92.58**
Unseen drug	Our loop	10	**91.65**	**92.58**
Unseen protein	Fixed model	1	76.40	78.75
Unseen protein	Direct	1	83.26	80.53
Unseen protein	Our loop	1	83.26	80.53

Endpoint	Method	Labs	AUROC	AP
Unseen protein	Fixed model	10	85.80	78.92
Unseen protein	Direct	10	**87.63**	**85.07**
Unseen protein	Our loop	10	**87.63**	**85.07**

Table A.1: Control comparison for the task-reference models. Each method is reported with one laboratory and with ten: comparing rows within a block isolates design search, and comparing blocks isolates laboratory participation. Direct and BioCoLoop attain the same primary score in every ten-laboratory run; their selected predictors need not be identical. Entries are held-out mean $\pm$ sample SD across completed training/search seeds, multiplied by 100. The data partitions remain fixed; this SD measures run-to-run variation, not dataset uncertainty. DTI uses the random held-out endpoint. Bold marks all displayed maxima.

		DTI	Proteomics	Cell perturbation		
Labs	Method	AUROC ↑	AP ↑	VCC Top-1 ↑	Norman Top-1 ↑	Tahoe Top-1 ↑
1	Fixed model	82.83±1.22	25.05±4.56	29.94±0.00	**25.56**±0.00	19.40±0.00
1	Direct	89.89±1.02	20.13±0.07	21.09±4.38	14.15±0.00	15.42±6.21
1	Our loop	89.89±1.02	20.13±0.07	20.22±2.22	14.15±0.00	15.42±6.21
10	Fixed model	91.80±2.55	**37.07**±0.08	**31.54**±0.08	15.48±0.00	**25.87**±1.72
10	Direct	**93.76**±1.11	36.99±0.22	**31.54**±0.08	22.25±0.45	22.39±2.59
10	Our loop	**93.76**±1.11	36.99±0.22	**31.54**±0.08	22.25±0.45	22.39±2.59
Completed seeds ($n/3$)		3/3	3/3	3/3	3/3	3/3

APPENDIX B. LABORATORY, SOURCE AND PROPOSER SENSITIVITY

Sensitivity study design. We vary laboratory participation, data sources, proposer and search budget using the main study's target scorers. Laboratory-count studies cover all five endpoints at $K \in \{1, 2, 5, 10\}$ (DTI: seed 61; proteomics/cell: seeds 61–63). Independent-source DTI/proteomics use seed 61; external-source VCC uses seeds 61–63. Proposal studies compare 12 designs/six slots on all endpoints (DTI: seed 42; others: 61) and 36 designs/24 slots on four lighter endpoints (seed 61), with 100 rounds per candidate. Budget effects are assessed within each protocol because the libraries differ. Main-table seeds 42–44 and protocols remain unchanged.

B.1 NUMBER OF LABORATORIES

Increasing participation adds the first K original laboratories and their training/development data. The fixed-pool study instead merges all ten training partitions into K groups while preserving equal-weight evaluation over the original ten development panels. Thus the first experiment adds evidence, whereas the second changes training partitions without changing the selection population. Both use the default design, 100 rounds and the same $K = 10$ fit. For cell tasks, repartitioning also changes within-group contrastive negatives and optimization steps. Zero between-seed SD can arise from identical single-batch updates to a fixed pretrained head; it describes training variability on these fixed partitions.

Table B.1: Increasing participation. A nested set of the original ten client partitions participates, so increasing K also adds training data. The default design and 100-round trainer are fixed. DTI uses seed 61; the four light endpoints report mean ± sample SD over seeds 61–63. Scores are metric $\times 100$; bold denotes the best K within each row, including displayed ties.

Task	Metric	$K = 1$	$K = 2$	$K = 5$	$K = 10$
DTI	AUROC	83.65	84.43	89.62	**93.88**
PTPC	AP	26.03 ± 5.57	37.38 ± 0.60	29.63 ± 2.50	**37.54 ± 0.27**
VCC	Macro Top-1	30.02 ± 0.78	**39.56 ± 0.11**	31.88 ± 0.00	31.48 ± 0.16
Norman	Macro Top-1	**25.56 ± 0.00**	18.22 ± 0.00	19.68 ± 1.11	15.48 ± 0.00
Tahoe	Macro Top-1	19.40 ± 0.00	25.37 ± 0.00	20.90 ± 0.00	**26.37 ± 0.86**

Table B.2: Fixed data pool. All target data are retained and merged into K clients; only the partition count changes. The default design and 100-round trainer are fixed. DTI uses seed 61; the four light endpoints report mean ± sample SD over seeds 61–63. Scores are metric $\times 100$; bold denotes the best K within each row, including displayed ties.

Task	Metric	$K = 1$	$K = 2$	$K = 5$	$K = 10$
DTI	AUROC	85.47	88.96	88.05	**93.88**
PTPC	AP	34.59 ± 5.66	37.36 ± 1.89	**39.19 ± 0.43**	37.54 ± 0.27
VCC	Macro Top-1	29.49 ± 0.00	28.80 ± 0.20	31.06 ± 0.22	**31.48 ± 0.16**
Norman	Macro Top-1	**49.63 ± 4.20**	19.56 ± 0.20	26.12 ± 0.11	15.48 ± 0.00
Tahoe	Macro Top-1	22.39 ± 0.00	21.39 ± 0.86	20.40 ± 0.86	**26.37 ± 0.86**

B.2 INDEPENDENT SOURCES AND TARGET EVALUATION

For DTI and cell transfer, the full target training pool and each auxiliary collection form separate laboratories, combined by sample-weighted aggregation (Table B.3). Target-only, individual-source and combined-source arms share initialization, original target-development panels for checkpoint selection, and the target test set.

BioSNAP, Davis and Human from the pinned TAPB release contribute 19,092, 1,721 and 2,929 DTI pairs after de-duplication against target splits and preceding sources; all five source arms use 100 rounds. VCC adds Replogle K562, Nadig HepG2 and Jiang IFNB/K562, retaining 60, 60 and 58 interventions (Replogle et al., 2022; Nadig et al., 2025; Jiang et al., 2025a). Source panels retain measured genes and exclude target development/test intervention identities from training; the source manifest records subsets, feature intersections and exclusions.

Table B.3: Source and context mapping in the cross-source experiments. The DTI units are database collections, the cell units preserve their study and biological scenario, and each proteomic auxiliary study owns one client and its response head. The ten PTPC clients remain partitions of one target corpus. Biological contexts within a study are not counted as additional independent studies.

Task	Client unit	Sources and biological contexts
DTI	Four data collections	Target BindingDB; BioSNAP; Davis; Human. Each collection contributes one client after pair-level de-duplication.
Cell	Four study/scenario clients	Target VCC; Replogle K562; Nadig HepG2; Jiang IFNB/K562. Each auxiliary source contributes one client.
Proteomics	Ten target partitions plus three source clients	Clients 0–9: PTPC partitions; client 10: Lin (A549, H292, PC9, PSC1); client 11: Ruprecht (A549, Calu6, Calu1, 2030, 2122); client 12: decryptE (Jurkat, five doses).

Table B.4: Independent-source transfer on the same target heldout set, including every prespecified source and their combination. DTI (seed 61) adds BioSNAP, Davis or Human; VCC (seeds 61–63, mean $\pm$ SD) adds Replogle, Nadig or Jiang. The PTPC auxiliary multi-task model (seed 61) adds Lin or Ruprecht and is compared only with its matched auxiliary target-only model. The two-source PTPC protocol shown here uses Lin and Ruprecht; the three-source extension is reported separately in Table B.6. Changes are paired against target-only training; bold marks the best score within a target/model block. Scores are multiplied by 100.

Target	Additional source	Metric	Score ↑	Δ (pp)
DTI	Target only	AUROC	85.47	+0.00
DTI	+ BioSNAP	AUROC	82.44	-3.03
DTI	+ Davis	AUROC	87.86	+2.39
DTI	+ Human	AUROC	**87.95**	+2.48
DTI	+ All three	AUROC	84.30	-1.18
VCC	Target only	Macro Top-1	29.49 $\pm$ 0.00	+0.00 $\pm$ 0.00
VCC	+ Replogle	Macro Top-1	27.12 $\pm$ 0.08	-2.37 $\pm$ 0.08
VCC	+ Nadig	Macro Top-1	26.94 $\pm$ 0.18	-2.55 $\pm$ 0.18
VCC	+ Jiang	Macro Top-1	**33.07 $\pm$ 0.00**	+3.58 $\pm$ 0.00
VCC	+ All three	Macro Top-1	27.33 $\pm$ 0.00	-2.15 $\pm$ 0.00
PTPC auxiliary	Target only	AP	34.98	+0.00
PTPC auxiliary	+ Lin	AP	35.01	+0.03
PTPC auxiliary	+ Ruprecht	AP	**35.50**	+0.52
PTPC auxiliary	+ Both	AP	35.31	+0.34

Table B.5: Fixed-design source–scenario ablation on VCC. K: training clients; N: training intervention conditions. The two $K = 4, N = 60$ arms share a 15-condition VCC anchor and differ in the sources of the other three clients. All arms use 100 rounds and the same target evaluation. Mean $\pm$ sample SD over three seeds; best means are bold. Cell counts and gene panels remain source-specific; protocol details are in Appendix B.

Training sources	K	N	Top-1 (%) ↑	MRR (%) ↑
VCC, limited data	1	15	29.36 ± 0.26	$\mathbf{56.97 \pm 0.19}$
VCC, all target data	1	60	29.43 ± 0.09	56.00 ± 0.09
VCC, same-source split	4	60	$\mathbf{30.70 \pm 0.42}$	56.24 ± 0.04
VCC + external studies	4	60	28.82 ± 0.00	56.30 ± 0.00

Table B.4 retains every source-specific gain and decrease. Across three cell-task seeds, Jiang raises Top-1 by 3.58 points (paired intervention-bootstrap 95% interval: $[-2.11, 9.51]$).

Matched source–scenario laboratories. Table B.5 compares 15 or 60 VCC interventions in one laboratory, the same 60 split across four laboratories, and four source-defined laboratories with 15 interventions each (VCC, Replogle/K562, Nadig/HepG2, Jiang/IFNB/K562). Both four-laboratory arms share a 15-intervention VCC anchor. Hash-based auxiliary sampling excludes target training/development/test identities and cross-source duplicates before fitting. Intervention counts match, but cell counts differ: 77,771 same-source versus 33,469 cross-scenario cells; source-specific gene panels and loss masks are retained. All arms use identical head initialization, the default design, 100 rounds, seeds 61–63 and no proposals. The original ten target-development panels select checkpoints; testing uses 20 interventions and 360 five-option queries. Independent rescoring verifies all twelve fits. Cross-scenario Top-1 is lower in every seed (mean difference -1.88 points); MRR differs little.

Two-source proteomic transfer and proposal search. Lin (Lin et al., 2026) and Ruprecht (Ruprecht et al., 2020) each add one study-level laboratory to the ten target partitions. Their continuous responses differ from the binary target endpoint. A shared masked-protein encoder of width 64 and drug encoder of width 32 therefore feed separate heads for target binary efficacy, Lin CellTiter-Glo response and Ruprecht potency intervals. Encoders use sample-weighted aggregation; each private head is updated only by its own source laboratories. This auxiliary adapter has its own target-only control, separate from the main PTPC predictor.

Lin provides 28 conditions (20 training/eight development, from five training and two development drugs), with 24-hour, 10-μM protein log-fold changes and the mean of three released CellTiter-Glo ratios; the HCT16/HCT116 alias is excluded from auxiliary training. Ruprecht provides 109 conditions (85 training/24 development, from 17 training and five development drugs), pairing 24-hour protein log-fold changes with $\log_{10}$-nM EC50: 49 exact, 18 approximate and 42 right-censored labels. The loss is squared distance to the potency interval; explicit lower bounds or values $\geq 10{,}000$ nM are censored, and approximate values receive factor-of-two intervals. Profiling dose is omitted from model inputs, although source acquisition doses relate to response. The target has 386 training and 105 development examples, with 148 test observations across 92 compounds and 22 positives.

Each source setting uses 100 rounds and an independent two-proposal Luna search. All 12 fits and eight valid calls complete; Fixed and Loop have identical rankings in all four settings. Their AP/AUROC are 34.98/74.75 (target only), 35.01/74.78 (+Lin), 35.50/74.28 (+Ruprecht) and 35.31/74.17 (+both). Paired compound-bootstrap 95% intervals (1,000 resamples) for the three AP changes are $[-0.20, 0.28]$, $[-1.46, 2.64]$ and $[-1.62, 2.49]$ points. These are source-transfer results without an additional history-search gain.

Three-source proteomic transfer. decryptE (Eckert et al., 2025) contributes Jurkat proteomes after 18 hours at 1, 10, 100, 1,000 and 10,000 nM, predicting continuous metabolic activity relative to DMSO. Inputs combine log-transformed normalized protein intensities, an observation mask, drug fingerprints and dose indicators. Compounds are aligned by published names/aliases and structures; proteins by UniProt identifier. Splits keep all doses of a chemical graph together, with training-only normalization and target development/test exclusions by canonical and stereo-agnostic identity.

Of 145 processed matrices, one is an additional Vincristine copy, leaving 144 compounds. Excluding 12 target overlaps, six unsupported names and two identity/structure conflicts yields 124 compounds/620 conditions: 495 training observations from 99 compounds and 125 development observations from 25 compounds. The source manifest preserves exclusions and annotations.

All four arms—target only, +decryptE, +Lin/Ruprecht and +all three—share a four-head model, seed-61 initialization, optimizer, sample-weighted shared-encoder aggregation and 100 rounds. Each source owns its private head. Equal-laboratory target-development AP selects checkpoints, breaking ties by binary cross-entropy; auxiliary development scores are diagnostic. These fixed-design arms use zero proposals.

Table B.6: Three-source proteomic transfer on PTPC. All four arms use the same four-head adapter, seed 61, 100-round fitting schedule and target-development checkpoint rule. AP and AUROC are multiplied by 100; bold identifies the best value in each metric. The target test set contains 148 observations from 92 compounds. These are fixed-design source comparisons; the separate two-source proposal pilot is described in Appendix B.2.

Additional source	AP ↑	AUROC ↑	Δ AP (pp)
Target only	34.98	**74.75**	+0.00
+ decryptE	35.03	74.21	+0.05
+ Lin + Ruprecht	35.31	74.17	+0.34
+ Lin + Ruprecht + decryptE	**35.68**	74.28	+0.70

All three sources raise AP from 34.98 to 35.68 (+0.70 points); adding decryptE to Lin/Ruprecht adds 0.37 AP and 0.11 AUROC points. Target-only retains the highest AUROC (74.75 versus 74.28). The four single-seed CPU fits take 101.21 seconds; independent rescoring of sealed predictions agrees exactly.

One study per training laboratory. Merging the 386 target training examples into one laboratory gives $K = 1, 2, 3, 4$ study-defined laboratories with 386, 881, 491 and 986 conditions, respectively. Lin, Ruprecht and decryptE each remain a separate laboratory. The four-head model, its initialization (shared by all arms within a seed), losses, aggregation and private-head ownership are unchanged. The target now executes seven minibatches per local epoch rather than aggregating ten partitions, requiring a separately trained target-only control. All four arms use 100 rounds, zero proposals and equal-weight checkpoint selection over the original ten target-development panels. Training participation therefore changes while the target evaluation population stays fixed. The original seed-61 run takes 76.18 CPU seconds; independently rescored AP/AUROC differ by at most 1.11×10^{-16}. Seeds 62 and 63 replicate it with only the seed changed, under an analysis fixed before they ran (all four arms, mean ± sample SD, paired differences from the target study); their four-source arm reuses the verified fixed-design fits of the same seeds. Because three seeds left the AP spread large, we then registered an eight-seed analysis (adding seeds 64–68, all arms fitted fresh, 100,000 seed-block bootstrap draws, win/loss counts, no exclusions) before fitting the new seeds. Every arm is independently rescored from saved predictions.

Over the eight seeds, all three auxiliaries raise AUROC from 72.06 to 74.13 (paired $+2.07$ points, 95% interval [0.65, 3.38]; seven of eight seeds higher) and AP from 34.60 to 36.82 ($+2.22$, [-1.19, 5.59]; four higher, four lower). Seeds 61–63 alone give $+5.97$ AP, and seed 64 gives $+8.65$; seeds 65–68 give -5.87 to -0.23, with seed 65's target-only model reaching an unusually high 41.71 AP. DecryptE alone gives $+2.09$ AUROC ([0.75, 3.33]; seven of eight) and $+2.79$ AP ([-0.64, 6.24]; six of eight). Lin and Ruprecht alone change AP by -0.30 and AUROC by $+0.18$, with intervals spanning zero. The target-only AP varies widely across seeds (SD 3.73), so the ranking benefit of the auxiliary studies is more stable than the precision-at-the-top benefit. These fixed-design comparisons vary training data and participating studies.

Within-study proteomic contexts. HCC1143, HCC1806 and MDA-MB-453-1 are three contexts within mtPTDS (Sun et al., 2026), each with 47 conditions and 3,501 target-overlapping proteins. Training excludes target development/test compounds. These runs use the original proteomic adapter and seeds 61–63.

Table B.7: One worker per proteomic study. The target training pool is merged into one client; Lin, Ruprecht and decryptE each form one additional client. K counts training clients and N counts training conditions. All four fixed-design arms use 100 rounds, a common initialization within each seed, and equal-weight evaluation over the original ten target development panels. Scores are mean $\pm$ sample SD over seeds 61–68 on the common target held-out set, in percent; Δ is the mean paired change from the target study alone. Seeds 64–68 were added under an analysis fixed before their fits, with fresh fits for every arm in those five seeds. Best means are bold.

Training sources	K	N	AP ↑	ΔAP	AUROC ↑	ΔAUROC
Target study	1	386	34.60 $\pm$3.73	–	72.06 $\pm$1.93	–
+ decryptE	2	881	**37.39** $\pm$2.64	+2.79	**74.14** $\pm$1.48	+2.09
+ Lin + Ruprecht	3	491	34.30 $\pm$4.88	−0.30	72.24 $\pm$2.03	+0.18
+ Lin + Ruprecht + decryptE	4	986	36.82 $\pm$2.90	+2.22	74.13 $\pm$1.74	+2.07

Table B.8: Within-study PTPC context transfer, reported separately from independent-source transfer. HCC1143, HCC1806 and MDA-MB-453-1 belong to the same mtPTDS study. Values are mean $\pm$ sample SD over seeds 61–63, multiplied by 100. Confidence intervals resample target compounds while retaining paired seed predictions. These contexts are not additional independent studies; the original model and results are unchanged.

Additional context	AP ↑	Δ AP (pp)	Paired 95% CI
Target only	34.59 ± 5.66	+0.00 ± 0.00	[0.00, 0.00]
+ HCC1143	34.94 ± 5.15	+0.35 ± 0.56	[-1.22, 1.60]
+ HCC1806	35.23 ± 5.35	+0.64 ± 0.32	[-0.78, 1.77]
+ MDA-MB-453-1	35.20 ± 5.31	+0.61 ± 0.44	[-0.85, 2.04]
+ All three	**35.47** ± **6.36**	+0.87 ± 8.89	[-2.44, 3.99]

B.3 Research-model and six-slot proposal comparison

The original twelve-design study uses Appendix A's direct/history protocol, shared first proposal, initialization and 100-round fits. Prefixes 0/1/2/4/6 are scored after search; repeated fits retain their logical budget charge. Both proposer models share task messages, design and fitting budgets. Appendix E.2 adds conventional controls; E.9 reports the separate expanded-recipe follow-up.

Across ten endpoint/backend pairs, Norman/Luna gains 0.81 Top-1 points, DTI/Luna loses 0.33 AUROC points and eight tie. Independent rescoring verifies all twenty DTI prefix scores on 5,505 test pairs and their checkpoint/output hashes. Figure 4 shows every prespecified development/test prefix.

B.4 Extended search and task-dependent saturation

Both backends and modes run 24 slots on PTPC, VCC, Norman and Tahoe using 36 designs: three learning rates, three weight decays, two coordinator-momentum values and two residual-head settings. Prefixes are 0, 1, 2, 4, 6, 8, 12, 16, 20 and 24. Initialization, fitting, selection and invalid-slot accounting follow B.3; DTI remains a six-slot comparison.

Luna completes 192/192 valid slots and Qwen 116/192. Twelve of 16 trajectories have no primary-score improvement in the last eight slots, although 11–24 designs remain unvisited. Table B.10 measures first crossing of the lower terminal development score in each pair. Norman/Luna history crosses in 4 versus 22 slots, while Tahoe/Luna direct crosses in 5 versus 13. All 24 slots still execute: these are retrospective trajectory differences, not realized compute savings. Table B.12 preserves the retained configurations, and Figure 5 includes the unfavorable VCC/Tahoe held-out trajectories diagnosed in E.3.

B.5 Executed designs and proposal traces

The short study uses D00–D11 (Appendix A.2); the extended study uses all $3 \times 3 \times 2 \times 2 = 36$ combinations below. Identifier $\mathtt{L}i\mathtt{W}j\mathtt{M}k\mathtt{R}l$ indexes each factor ($i, j \in \{0, 1, 2\}$; $k, l \in \{0, 1\}$). Thus

Table B.9: Original and extended proposal searches, shown together without mixing configuration libraries. D and H denote direct and history-guided search, and @ denotes the number of proposal slots. Every candidate receives 100 training rounds using the same ten laboratories. Panel (a) uses DTI seed 42 and other-task seed 61; panel (b) uses seed 61. The six-slot prefixes in the two panels belong to different searches. DTI uses AUROC, proteomics uses average precision, and cell tasks use Top-1, all multiplied by 100. Bold denotes the highest displayed score in each row, including ties; dashes indicate runs not performed.

Task	Proposer	Fixed	D@6	H@6	D@24	H@24
(a) 12 designs, six proposal slots						
DTI	Luna	94.27	**94.60**	94.27	–	–
DTI	Qwen	**94.27**	**94.27**	**94.27**	–	–
Proteomics	Luna	**37.27**	**37.27**	**37.27**	–	–
Proteomics	Qwen	**37.27**	**37.27**	**37.27**	–	–
VCC	Luna	**31.50**	**31.50**	**31.50**	–	–
VCC	Qwen	**31.50**	**31.50**	**31.50**	–	–
Norman	Luna	15.48	22.30	**23.11**	–	–
Norman	Qwen	15.48	**22.30**	**22.30**	–	–
Tahoe	Luna	25.37	**26.87**	**26.87**	–	–
Tahoe	Qwen	**25.37**	**25.37**	**25.37**	–	–
(b) 36 designs, 24 proposal slots						
Proteomics	Luna	**37.27**	**37.27**	**37.27**	**37.27**	**37.27**
Proteomics	Qwen	**37.27**	**37.27**	**37.27**	**37.27**	**37.27**
VCC	Luna	**31.50**	16.94	16.94	20.23	20.01
VCC	Qwen	**31.50**	**31.50**	**31.50**	19.61	19.61
Norman	Luna	15.48	**35.48**	23.11	23.11	23.11
Norman	Qwen	15.48	**22.30**	**22.30**	**22.30**	**22.30**
Tahoe	Luna	25.37	13.43	**26.87**	13.43	13.43
Tahoe	Qwen	25.37	**26.87**	**26.87**	13.43	13.43

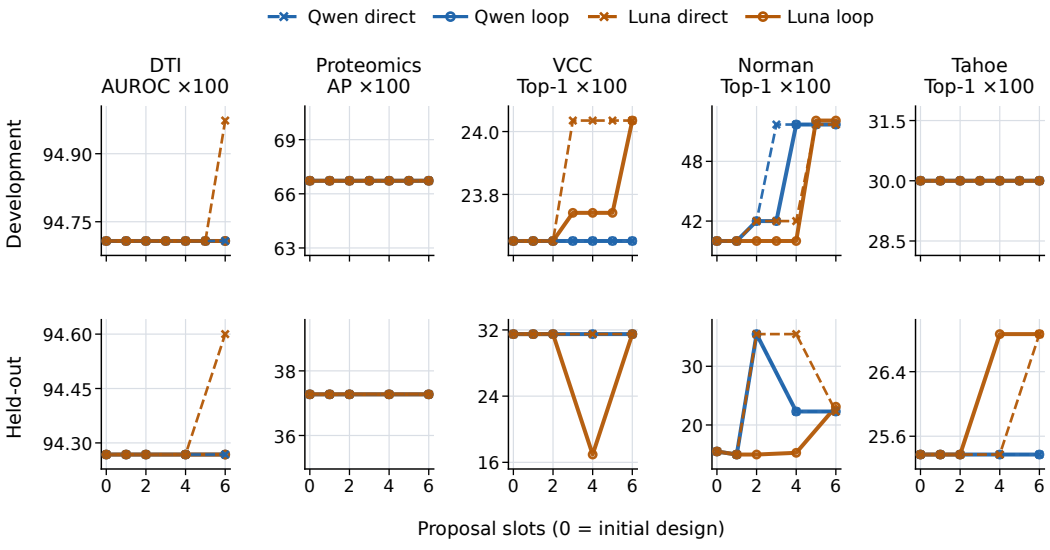


Figure 4: Research trajectories with and without evidence history in the six-slot protocol. Curves report development selection and held-out scores at each proposal prefix, where each candidate receives 100 rounds and slot 0 is the initial design.

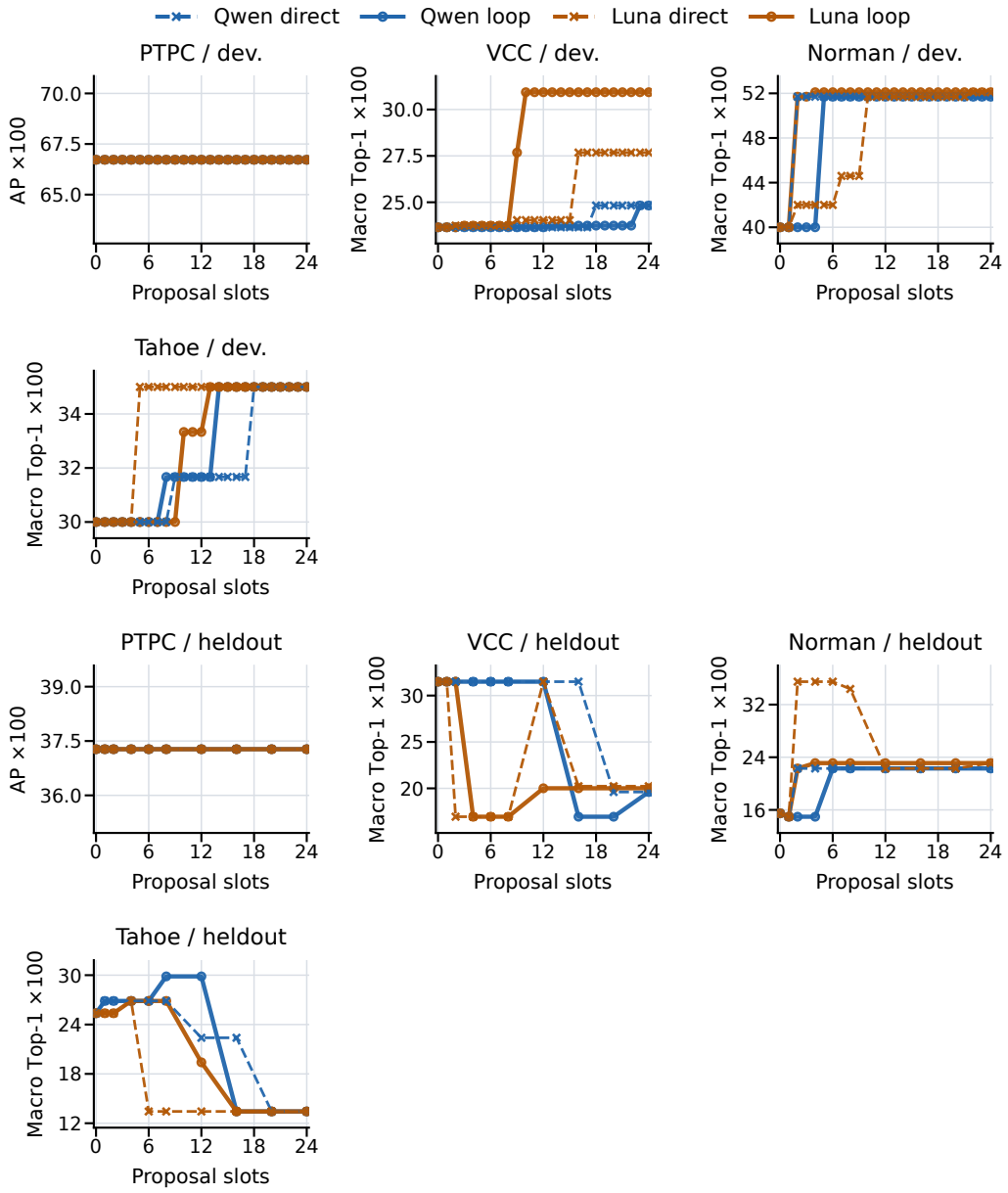


Figure 5: Development selection (upper panels) and held-out performance (lower panels) in the separate 36-design, 24-slot protocol on four endpoints. Curves are paired within the same backend, initialization and design space. Score plateaus may still include loss-tiebreak selections. All prespecified prefixes are displayed. Six-slot points here belong to this 36-design protocol, not the separate 12-design experiment.

Table B.10: Retrospective development-trajectory diagnostics for the two proposal-search protocols. Each D/H pair gives direct/history. The target is the lower terminal development score of the two runs; First is its earliest attainment, with slot 0 denoting initialization. Last gain records the final strict primary-score increase; Last retain also includes loss-tiebreak selections. All six or 24 slots were executed: these target crossings are descriptive and did not save training budget. Valid counts report accepted proposal formats, not retained model improvements.

Task	Proposer	Dev. target	First D/H	Last gain D/H	Last retain D/H	Valid D/H
(a) 12 designs, six proposal slots						
DTI	Luna	94.71	0/0	6/0	6/0	6/6
DTI	Qwen	94.71	0/0	0/0	0/0	6/6
Proteomics	Luna	66.72	0/0	0/0	0/0	6/6
Proteomics	Qwen	66.72	0/0	0/0	0/0	6/5
VCC	Luna	24.04	3/6	3/6	3/6	6/6
VCC	Qwen	23.65	0/0	0/0	0/0	6/6
Norman	Luna	51.70	5/5	5/5	5/5	6/6
Norman	Qwen	51.70	3/4	3/4	3/4	6/6
Tahoe	Luna	30.00	0/0	0/0	6/3	6/6
Tahoe	Qwen	30.00	0/0	0/0	0/0	4/4
(b) 36 designs, 24 proposal slots						
Proteomics	Luna	66.72	0/0	0/0	0/0	24/24
Proteomics	Qwen	66.72	0/0	0/0	0/0	19/14
VCC	Luna	27.68	16/9	16/10	16/10	24/24
VCC	Qwen	24.83	18/23	18/23	18/23	11/16
Norman	Luna	52.10	22/4	22/4	22/7	24/24
Norman	Qwen	51.70	2/5	2/5	22/19	15/12
Tahoe	Luna	35.00	5/13	5/13	5/13	24/24
Tahoe	Qwen	35.00	18/14	18/14	18/14	14/15

`L0W1M0R1` means learning rate 10^{-4}, weight decay 10^{-3}, zero coordinator momentum and an enabled residual head. Proximal coefficients are zero; task interfaces stay fixed.

Table B.11: Exact factor-to-index mapping for the extended 36-design library. Identifiers concatenate the four prefix–index pairs in L, W, M, R order, and all combinations of the listed values are included.

Factor	Identifier prefix	Index 0	Index 1	Index 2
Learning rate	`L`	10^{-4}	10^{-3}	10^{-2}
Weight decay	`W`	0	10^{-3}	10^{-2}
Coordinator momentum	`M`	0	0.9	–
Residual module	`R`	disabled	enabled	–

The reproducibility archive retains all 504 attempted slots (120 short, 384 extended), including acceptance decisions, configurations, incumbent-relative changes and measured development scores.

Table B.12: All retained proposals for the Norman/Luna extended 36-design, 24-slot study, seed 61. The two modes share initialization and first proposal. lr, wd, m and R denote learning rate, weight decay, server momentum and residual-head indicator. Development selection is followed by evaluation at every prespecified heldout prefix; the full attempted-slot trace is retained separately.

Mode	Slot	Executed configuration	Dev. Top-1
Direct	0	Common initial recipe	40.00
Direct	1	lr=0.001, wd=0.001, m=0, R=1	40.00
Direct	2	lr=0.001, wd=0.001, m=0.9, R=1	42.00
Direct	7	lr=0.001, wd=0.01, m=0.9, R=1	44.60
Direct	10	lr=0.0001, wd=0, m=0, R=1	51.70
Direct	11	lr=0.0001, wd=0.001, m=0, R=1	51.70
Direct	22	lr=0.0001, wd=0.001, m=0, R=0	52.10
Loop	0	Common initial recipe	40.00
Loop	1	lr=0.001, wd=0.001, m=0, R=1	40.00
Loop	2	lr=0.0001, wd=0.001, m=0, R=1	51.70
Loop	4	lr=0.0001, wd=0.001, m=0, R=0	52.10
Loop	7	lr=0.0001, wd=0, m=0, R=0	52.10

APPENDIX C. PUBLIC SELF-IMPROVING-AGENT COMPARISON

C.1 SHARED BIOLOGICAL TASK INTERFACE

AI-Scientist-v2 (Yamada et al., 2025) and AI-Researcher (Tang et al., 2025) use BioCoLoop's fitting and evaluation service, at source versions `96bd51617cfd` and `f9a6f8480860`, respectively (SakanaAI, 2025; HKUDS, 2025). Both use the registered executable design space of twelve configurations, at most six proposals, 100 rounds per valid candidate and the shared held-out scorer. The service fits D00 afresh, then executes submitted `design_id` values; the hypothesis, experiment and expected-effect fields record rationale. Repeated or invalid requests consume slots, and agents may stop early. The original development score/loss rule selects checkpoints.

AI-Scientist-v2 retains its experiment manager, tree-search paths, reviews and journal. AI-Researcher retains native idea selection, planning, implementation, review handoffs and iterative analysis. Prompts use the registered design interface; arbitrary generated-code execution, external retrieval, figure review and manuscript generation are disabled. The matched blocks use local Qwen2.5-7B-Instruct (temperature 0.7, top-p 0.9, at most 512 generated tokens per call) and GPT-5.6 Luna (low reasoning effort, text-only bridge, tools disabled). Adapters allow at most 160 model calls and 28,000 input tokens per call, raised to 65,536 for the four starred Luna AI-Researcher retries (C.2). Qwen uses generation seeds; Luna labels identify training/search repetitions. Per-run call counts, token usage and transport receipts document the variable language-model costs.

C.2 PREDICTION QUALITY AND EXECUTION OUTCOMES

A scored run completes its agent's research process within the registered cap, seals development selection and passes independent held-out rescoring. Table 1 summarizes seeds 42–44; fixed, direct and BioCoLoop rows are complete. Daggers mark partial cells and dashes mark unscored cells, without imputation. Partial cells are Qwen2.5-7B-Instruct AI-Researcher on VCC ($2/3$ seeds, F_{tool}) and GPT-5.6 Luna AI-Scientist-v2 on Tahoe ($2/3$, F_{svc}).

Four Luna AI-Researcher cells exceeded the 28,000-token admission (F_{ctx}) and were re-run at 65,536 tokens with the same tasks, seeds, design space, six-proposal cap, training and scorer. Their completed cells carry stars, with per-seed records retained. BindingDB (DTI), the longest native refinement trajectory, remains unscored ($0/3$): one frozen seed reached 29,167 input tokens (F_{ctx}), and two terminated in service transport (F_{svc}). Extended-admission retries also ended in transport before held-out selection was sealed. The revised table reports laboratory access instead of averaging ranks over different completed endpoint sets.

Receipts distinguish native pipeline (F_{pipe}), tool-transport (F_{tool}), registered context-limit (F_{ctx}) and Luna service-transport (F_{svc}) terminations before sealed selection. An earlier seed-42 Qwen AI-Scientist-v2 DTI attempt trained three candidates before substage schema validation failed. That historical attempt is distinct from the completed run represented in the table. Execution failures describe this adaptation and infrastructure, not predictive scores or evidence that the original framework cannot solve the task.

Held-out scoring follows sealed development selection. An independent verifier recomputes each scored cell's metric from saved predictions and checks checkpoint and artifact hashes. DTI discrepancies are at most 10^{-15}. Records retain raw calls, native stage history, candidate requests, selections and verification receipts.

C.3 TASK-TRANSPORT REPAIR

The AI-Researcher adaptation removes repeated task packets from stage prompts, preserving the complete task and evidence in system prompts, native reasoning and tool results. The bridge accepts an unambiguous `parameters` alias for `arguments` and standard single-tool-call envelopes without inferring actions or values. This revision preserves model weights, context allowance, training and scoring. It applies across all five endpoints and the Luna sensitivity; failures remain recorded.

C.4 TEN-LABORATORY PILOT

Table 1 gives the public agents one laboratory. An earlier seed-42 pilot instead connected the Qwen controllers to all ten laboratories through the same interface, design menu, six-proposal cap and scorer. AI-Scientist-v2 reaches 95.07 DTI AUROC versus 94.60 for BioCoLoop and the same scores on proteomics, VCC, Norman and Tahoe (37.12, 31.63, 22.74 and 23.88), where it selects the same designs. The repaired AI-Researcher adapter matches BioCoLoop on proteomics, VCC and Norman; its DTI and Tahoe runs remain unscored. This single-seed pilot was not extended to three seeds, so it is not averaged into the main table. It shows that the collaborative interface is not tied to BioCoLoop's controller: with ten laboratories, an independent agent reaches comparable models.

APPENDIX D. SINGLE-LABORATORY CONFIGURATION-SEARCH CONTROLS

D.1 SEARCH PROTOCOL AND PRIMARY RESULTS

We add conventional configuration-search controls to test whether more extensive search within one laboratory can match the collaborative framework. All new controls access only laboratory 0's training and development partitions. They use the same twelve registered designs, task-specific initialization, original GPU training implementation and held-out scorers as the main study. Each design is fitted afresh for 100 rounds at seeds 42–44. Development score, followed by development loss, selects the design and checkpoint.

Random search evaluates D00 and six distinct randomly sampled designs. Bayesian optimization evaluates D00, the first two proposals from the corresponding seeded random-search sequence, and four proposals selected by Gaussian-process expected improvement. Its fixed Matérn-$5/2$ kernel uses unit length scales, diagonal regularization $\alpha = 10^{-6}$, normalized outcomes and an improvement offset of 0.01. Design features encode log-scaled learning rate, ordinal weight decay, normalized momentum and the binary residual flag. Both methods have a nominal budget of 700 candidate-rounds. Exhaustive enumeration evaluates all twelve registered configurations, using 1,200 candidate-rounds; this covers the registered menu rather than a full Cartesian hyperparameter grid. The search decisions are replayed using only the development results of queried configurations from the freshly trained, fixed-seed candidate cache. The BioCoLoop rows reuse the original Qwen2.5-7B-Instruct runs, with and without collaborative participation.

Table D.1: **Single-laboratory search controls.** Mean ± sample SD over three training seeds, in percentage units. Search controls use only lab 0; the history-enabled BioCoLoop references use Qwen2.5-7B-Instruct with one or ten laboratories. The ten-laboratory setting adds training and evaluation evidence. The ten-laboratory fixed model is the D00 arm of the same six-arm study (Table A.1). Bold marks the highest displayed mean per column.

Method	Labs	DTI	Proteomics	Cell perturbation		
		AUROC↑	AP↑	VCC Top-1↑	Norman Top-1↑	Tahoe Top-1↑
LLM-free controls						
Fixed model	1	82.83 ±1.22	25.05 ±4.56	29.94 ±0.00	**25.56** ±0.00	19.40 ±0.00
Fixed model	10	91.80 ±2.55	**37.07** ±0.08	**31.54** ±0.08	15.48 ±0.00	**25.87** ±1.72
Random@6	1	90.10 ±0.71	20.13 ±0.06	21.10 ±4.41	14.10 ±0.04	17.91 ±3.95
Bayesian@6	1	90.10 ±0.71	20.01 ±0.24	18.61 ±2.78	14.12 ±0.04	16.92 ±4.80
12-design menu	1	90.10 ±0.71	20.13 ±0.06	18.61 ±2.78	14.12 ±0.04	15.42 ±6.21
History-enabled BioCoLoop references						
BioCoLoop	1	89.89 ±1.02	20.13 ±0.07	20.22 ±2.22	14.15 ±0.00	15.42 ±6.21
BioCoLoop	10	**93.76** ±1.11	36.99 ±0.22	**31.54** ±0.08	22.25 ±0.45	22.39 ±2.59

The ten-laboratory framework exceeds all three single-laboratory search strategies in mean score on all five endpoints. With one laboratory, every search strategy, including BioCoLoop, selects designs on laboratory 0's development panel alone. These designs differ across strategies and seeds but score similarly on the test set, and on proteomics and the cell tasks they fall below the single-laboratory fixed model: rankings on one laboratory's small development panel do not transfer reliably to the held-out set. Ten laboratories supply both more training data and a larger development population.

The two fixed-model rows separate these sources. Training D00 on all ten laboratories raises DTI, proteomics, VCC and Tahoe but lowers Norman from 25.56 to 15.48. Ten-laboratory BioCoLoop adds 1.96 AUROC on DTI and 6.77 Top-1 points on Norman, nearly matches the fixed model on proteomics (36.99 versus 37.07 AP) and ties it on VCC, and is 3.48 points lower on Tahoe; Appendix E.3 analyzes this development–test divergence. These rows compare access scenarios of complete workflows; Appendix A and Section 4.7 separately hold laboratory access fixed.

D.2 Search variability and held-out uncertainty

Table D.2 integrates random choices over all $\binom{11}{6} = 462$ subsets per training seed, always including D00. Ties integrate proposal orders under the original selection rule. These are expected selected-model scores, not ensembled predictions. The collaborative framework also exceeds these expectations on all five endpoints.

Table D.2: **Integrating random-search variability.** Expected Random@6 score over all $\binom{11}{6} = 462$ proposal subsets, integrating proposal-order ties. This averages selected-model metrics, not their predictions. Scores are percentages; the final row is the mean ± sample SD across three trained seeds after search randomness is integrated out.

Training seed	DTI AUROC↑	Proteomics AP↑	VCC Top-1↑	Norman Top-1↑	Tahoe Top-1↑
42	88.67	20.42	22.53	15.35	14.38
43	90.08	20.01	19.41	15.33	12.74
44	89.53	20.91	20.82	15.35	19.68
Mean ± SD	89.43 ±0.71	20.44 ±0.45	20.92 ±1.56	15.34 ±0.01	15.60 ±3.63

For these additional controls, a separate paired bootstrap uses 10,000 resamples of compounds for proteomics and interventions for the cell tasks. Each resample is shared across methods and the three trained models, and differences are averaged across seeds. These intervals quantify test-group uncertainty conditional on the trained models, complementing the seed standard deviations. The BioCoLoop-minus-random-search 95% intervals are [1.91, 29.97] AP points for proteomics and [−0.48, 22.49], [−9.53, 26.47] and [−3.98, 12.94] Top-1 points for VCC, Norman and Tahoe. The cell intervals span zero despite positive mean differences. These are descriptive intervals without multiplicity adjustment; the original study's uncertainty protocol remains as specified in Appendix A.5.

D.3 Generalization to unseen DTI entities

The same development-selected checkpoints are evaluated on the original random, unseen-drug and unseen-protein test sets. Exact input-identity checks confirm that none contains a training drug–protein pair; the unseen-drug and unseen-protein sets also have no exact drug-string or protein-sequence overlap with training, respectively. Scores are independently recomputed from saved, row-identified predictions.

Table D.3: **Single-laboratory search across three drug–target test splits.** AUROC and average precision (AP), mean ± sample SD over the same three seeds, expressed as percentages. Each method's development-selected predictor is reused across all three test splits. Lab access, candidate budgets and the historical Qwen BioCoLoop references are as in Table D.1. Bold marks the highest displayed mean per column.

Method	Labs	Random split		Unseen drug		Unseen protein	
		AUROC↑	AP↑	AUROC↑	AP↑	AUROC↑	AP↑
Fixed model	1	82.83 ±1.22	81.57 ±1.54	81.96 ±1.55	85.78 ±1.41	76.40 ±5.85	78.75 ±2.04
Random@6	1	90.10 ±0.71	88.06 ±1.38	89.07 ±0.81	90.19 ±1.38	83.70 ±2.12	81.60 ±4.27
Bayesian@6	1	90.10 ±0.71	88.06 ±1.38	89.07 ±0.81	90.19 ±1.38	83.70 ±2.12	81.60 ±4.27
12-design menu	1	90.10 ±0.71	88.06 ±1.38	89.07 ±0.81	90.19 ±1.38	83.70 ±2.12	81.60 ±4.27
BioCoLoop	1	89.89 ±1.02	87.77 ±1.79	88.39 ±1.66	89.35 ±2.34	83.26 ±2.61	80.53 ±5.10
BioCoLoop	10	**93.76** ±1.11	**92.73** ±1.81	**91.65** ±1.64	**92.58** ±2.16	**87.63** ±2.50	**85.07** ±5.34

BioCoLoop attains the highest mean AUROC and AP on all three DTI test sets. Compared with the three search controls, its AUROC gains are 3.66 points on the random split, 2.57 on unseen drugs and 3.93 on unseen proteins. The benefit therefore extends beyond the primary random-split endpoint.

The additional study comprises 180 complete candidate fits and 18,000 candidate-rounds. Their recorded fit times sum to 13.02 GPU-hours, excluding feature preparation, held-out scoring and statistical analysis. This is the cost of constructing the shared evaluation cache, rather than the cost of a single search. Aggregate per-seed scores, search definitions and source hashes accompany the tables in `tables/single_lab_search/`.

APPENDIX E. SEARCH CONTROLS, ALLOCATION AND VALIDATION

E.1 ORIGINAL MATCHED-ACCESS SEARCH PROTOCOL

Every method uses ten training laboratories, the same development panels, twelve designs, initialization and checkpoint rule, with six slots after D00 and 100 rounds per valid fit. The study reuses complete original candidate caches and Qwen/Luna trajectories (DTI seed 42; other endpoints seed 61), adding search controls rather than independent training repetitions. It is separate from the three-seed access comparison and the expanded-recipe protocol in E.9.

E.2 EXACT RANDOM SEARCH AND CONVENTIONAL OPTIMIZATION

Random search fits D00 and six distinct untried designs. Its expectation integrates all $\binom{11}{6} = 462$ subsets, including the possible encounter orders of score/loss-tied candidates. A shared-first control fixes the language model's first proposal and integrates $\binom{10}{5} = 252$ remaining subsets; an invalid first Qwen/Tahoe slot instead leaves five draws from eleven designs. GP–EI follows Appendix D's fixed kernel and acquisition rule. Selections are sealed before new held-out scoring. Random means average scores of development-selected single predictors, with SD describing policy randomness on the fixed cache.

The complete table retains the positive Norman/Luna comparison and the tied or unfavorable endpoints. Failed requests remain charged: Qwen history uses 600 fitted rounds on protein, and both Qwen modes use 500 on Tahoe, against a 700-round cap. Realized-fit matching gives Tahoe random 26.30 versus Qwen 25.37; fixing the valid first proposal gives DTI random 94.43 versus history 94.27. Exact probabilities and all sensitivity controls remain in the aggregate record.

E.3 SELECTION-STABILITY AUDIT

The original extended-search audit reconstructs 14 selected checkpoints, 140 development panels and all 80 held-out prefix scores without refitting or reselection. The development and held-out interventions are disjoint, with fixed five-option rosters sampled separately within each split.

VCC panels each contain two development interventions and Tahoe panels six. Equal-panel macro Top-1 therefore equals pooled intervention-macro Top-1 exactly; response-query counts cannot explain the decreases. The finite development populations (20 VCC interventions/302 queries and 60 Tahoe compounds) can expose repeated-selection sensitivity or biological shift; the original trajectories do not distinguish these explanations.

We additionally freeze 20 development-roster seeds (2026110900–2026110919), redraw four distractors and shuffle option order within each intervention's original pool, and score the initial and already selected final checkpoints on the same rosters. This diagnostic neither retrains nor reselects a model and uses no test responses.

Qwen's original advantage survives in 6/20 VCC and 7/20 Tahoe rosters; Luna/VCC remains positive on average, with 12/20 positive draws for each mode. These are roster draws on fixed interventions, not biological replicates; Appendix E.6 adds a separate gene-disjoint benchmark across study–cell contexts.

Retrospective stability-rule replay. To test whether a stricter acceptance rule mitigates development–test divergence, we replay the original fixed proposal traces of the main ten-laboratory runs (five tasks, seeds 42–44) under three rules frozen before this replay's held-out rescoring; the test sets were exposed previously. Each rule uses only per-laboratory development results. The SD rule accepts a replacement only if its pooled development gain exceeds the across-laboratory SD of that gain. The majority rule requires improvement on at least six of ten laboratories. The split-half rule requires improvement on both laboratory halves (0–4 and 5–9). Selections are sealed before held-out scoring, and no model is retrained. All three rules keep D00 on Tahoe, recovering 3.48 Top-1 points on average, but also reject D08 on Norman, losing 6.77 points; the rules also raise one proteomics seed by 0.24 AP, and the SD rule lowers DTI by 0.29 AUROC on average. Across the 15 task–seed cells, the majority and split-half rules each give three wins, nine ties and three losses (mean -0.64 points; seed-block bootstrap 95% interval $[-1.28, 0.02]$), and the SD rule gives three

Table E.1: **Matched-access configuration search.** All methods use the same ten laboratories, development panels and 12-design library. Searches receive six proposal slots after the common initialization, with up to 700 aggregation rounds. Scores are multiplied by 100. Random search reports its exact expectation and search-policy standard deviation, integrating all 462 six-element subsets; this is not variation across training seeds.

Method	DTI AUROC↑	Proteomics AP↑	VCC Top-1↑	Norman Top-1↑	Tahoe Top-1↑
Fixed model	94.27	**37.27**	**31.50**	15.48	25.37
Random search, no replacement	94.45 ±0.17	**37.27** ±0.00	27.53 ±6.48	22.66 ±3.41	26.59 ±0.58
GP–EI Bayesian optimization	**94.60**	**37.27**	**31.50**	22.30	25.37
Qwen: direct search	94.27	**37.27**	**31.50**	22.30	25.37
Qwen: history-guided search	94.27	**37.27**	**31.50**	22.30	25.37
Luna: direct search	**94.60**	**37.27**	**31.50**	22.30	**26.87**
Luna: history-guided search	94.27	**37.27**	**31.50**	**23.11**	**26.87**

One frozen training seed per endpoint (42 for DTI; 61 otherwise); agent and GP–EI rows each represent one search trajectory. The fixed reference uses 100 rounds. Failed proposals retain their slots: Qwen history uses 600 fitted rounds on proteomics, and both Qwen arms use 500 on Tahoe; other searches use 700. Matched-realized-fit and shared-first-proposal sensitivities are reported separately. Selection uses development scores only; the comparison reuses checkpoint-identical, fully trained candidates and previously exposed held-out panels. Bold marks the largest displayed mean, not statistical significance.

Table E.2: Development–test divergence in the extended search (training seed 61). Each pair is initialization → 24 proposals. CPU re-evaluation reproduces the original development scores; test scores are independently recomputed from archived query ranks. The equal-laboratory development mean equals the pooled intervention-macro mean because each laboratory contains the same number of development interventions. All scores are macro Top-1 percentages.

Task	Proposer / search	Development	Held-out
VCC	Qwen / direct and history	23.65 → 24.83	31.50 → 19.61
VCC	Luna / direct	23.65 → 27.68	31.50 → 20.23
VCC	Luna / history	23.65 → 30.93	31.50 → 20.01
Tahoe	Both / direct and history	30.00 → 35.00	25.37 → 13.43

wins, seven ties and five losses (mean -0.70). Stricter acceptance therefore trades the Tahoe gain against the Norman gain and gives no net improvement. We report this as a characterization of the trade-off and retain the original rule; testing larger development populations is a natural next step for stabilizing selection.

E.4 History under source heterogeneity

Using the source-defined protein/cell adapters in Appendix B.2, we hold source access fixed and compare direct, history and enriched-history proposals on twelve-design caches at three seeds. These controls isolate search from source-transfer gains; their fixed references are separate from the seed-61 transfer fits.

Qwen history is below direct on both tasks. Luna's enriched protein context scores 35.39 AP versus 34.94 for ordinary history and 35.35 for direct; cell modes select identical checkpoints within each seed. The enrichment changes several prompt fields and token costs. All 72 candidate predictions are reverified; these are new proposals over existing fits. The old adapters preserve the primary mean but lose panel disagreement before feedback. E.8 audits the versioned repair without replacing these search outcomes.

E.5 Training-promotion controls

A fresh $4 \times 3 \times 3 \times 10 = 360$-fit cache covers four tasks, three seeds (53–55), three training durations and ten designs. It compares uniform fitting with trajectory promotion, global score-only promotion, learning-rate-stratified score-only promotion and random promotion. Promotion uses ten 20-round

Table E.3: Development-roster sensitivity of the already selected initial and 24-proposal final checkpoints. Twenty frozen seeds resample four incorrect options within the same development-intervention pool; each comparison uses paired rosters. Entries are final-minus-initial macro Top-1 differences in percentage points. The standard deviation is across roster draws, not training seeds or new biological splits. No model is trained or reselected.

Task	Proposer / search	Original roster	New rosters	Positive draws
VCC	Qwen / both	$+1.17$	-1.34 ± 7.37	$6/20$
VCC	Luna / direct	$+4.03$	$+2.24 \pm 8.15$	$12/20$
VCC	Luna / history	$+7.28$	$+2.89 \pm 7.89$	$12/20$
Tahoe	Both / both	$+5.00$	-1.33 ± 4.82	$7/20$

Table E.4: Search with source-defined laboratories. Protein: PTPC, Lin, Ruprecht and decryptE; cell: VCC, Replogle, Nadig and Jiang. All methods share four training sources and the target development panels. Searches use twelve designs, six proposal slots and at most 700 rounds; the fixed reference uses 100 rounds. Mean $\pm$ sample SD over training/search seeds 61–63; scores $\times 100$. Random reports the exact policy expectation per seed, then its between-seed SD. Luna uses the same checkpoint cache but new proposal calls.

Method	Proteomic AP ↑	Cell macro Top-1 ↑
Fixed design	38.72 ± 1.94	28.82 ± 0.00
Random, exact expectation	36.37 ± 1.27	24.83 ± 4.23
Qwen direct	37.83 ± 1.28	22.36 ± 7.20
Qwen history	34.31 ± 4.96	19.41 ± 2.11
Luna direct	35.35 ± 3.62	22.88 ± 6.75
Luna history	34.94 ± 3.52	22.88 ± 6.75
Luna enriched history	35.39 ± 3.67	22.88 ± 6.75

screens plus six fresh 100-round fits; uniform trains all ten designs for 80 rounds. Both total 800 rounds and 160 aggregate development evaluations per search. Cache construction uses 24,000 actual rounds, distinct from a policy's logical budget.

Table E.5: Fresh-runtime allocation controls. Scores $\times 100$ are mean $\pm$ sample SD over training seeds 53–55. All policies use the same ten laboratories and ten designs, with 800 aggregation rounds and 160 aggregate development evaluations. Promotion policies screen ten designs for 20 rounds and restart six for 100; uniform allocation fits all ten for 80. Random promotion integrates all 210 six-design subsets per seed, and its displayed SD is across seed-conditional expectations. All 360 fits are newly trained; the original Table 2 is retained separately. Bold marks the best unrounded mean, including ties.

Allocation policy	Proteomics AP ↑	VCC Top-1 ↑	Norman Top-1 ↑	Tahoe Top-1 ↑
Uniform (80 rounds)	$\mathbf{37.47} \pm 0.55$	20.01 ± 0.00	11.48 ± 0.52	20.90 ± 2.99
Random promotion	37.37 ± 0.48	$\mathbf{22.49} \pm 1.70$	$\mathbf{22.63} \pm 0.64$	21.95 ± 0.04
Score only (global)	$\mathbf{37.47} \pm 0.55$	20.01 ± 0.00	21.70 ± 7.19	20.90 ± 0.00
Score only (LR groups)	$\mathbf{37.47} \pm 0.55$	20.01 ± 0.00	21.70 ± 7.19	20.90 ± 0.00
Score + trajectory	$\mathbf{37.47} \pm 0.55$	20.01 ± 0.00	22.32 ± 0.43	$\mathbf{22.89} \pm 1.72$

Development score/loss determines screening and final selection, with original slate order breaking complete ties. Random promotion integrates $\binom{10}{6} = 210$ subsets; its displayed SD is over three seed-conditional expectations. The fresh fits reproduce the original uniform/trajectory means. Trajectory promotion yields 4/8/0 wins/ties/losses against uniform, 4/7/1 against either score-only rule and 6/0/6 against random. Both score-only rules have identical task means. Selective longer fitting explains the broad gain over uniform; the promotion-rule advantage depends on the task.

E.6 History representation on a gene-disjoint cell benchmark

Replogle and Nadig supply four study–cell contexts (K562, RPE1, HepG2 and Jurkat), each acting as a training/evaluation laboratory. After vocabulary and measurement filtering, train/development/test contain 3,806/1,247/1,282 conditions and 1,208/414/406 perturbation genes. Both gene names and vocabulary identifiers are disjoint across partitions. This benchmark is separate from the original VCC, Norman and Tahoe endpoints.

Predictor and scoring. A corrected frozen scDEBART encoder predicts perturbed-minus-control mean log-normalized expression. The 512 response genes are selected by training variance and must be measured in all contexts. Five fixed rosters identify the true perturbation among five candidates per gene–context; every option excludes the same candidate-perturbation genes from scoring. Top-1 averages rosters within a gene, genes within a context, then contexts equally. Repeated rosters are not biological replicates.

Candidates and feedback. The 16 designs combine four learning rates ($10^{-5}, 3\times10^{-5}, 10^{-4}, 3\times10^{-4}$), two heads (fitted pretrained head or frozen head plus zero-output residual) and two objectives (MSE or MSE plus 0.1 within-batch contrastive loss). The original study uses seeds 61–63 (48 fits) and the extension adds seeds 64–68 (128 fits in total); all fits use 20 full-client rounds, batch size 32, weight decay 0.001 and zero coordinator momentum. Rounds 2/5/10/20 are selected by Top-1, lower cosine-ranking cross-entropy, then earlier round. D00 is the fully fitted 10^{-4}/MSE reference; round zero is diagnostic.

Qwen makes six requests after D00, with the first shared by direct/raw/structured modes. Direct receives identifiers without scores; raw and structured receive identical queried development facts. Structured cards organize these around reference designs, exact changes and expected-versus-observed outcomes. Proposals include design/reference identifiers, changed fields, rationale and a numeric prediction; executable fields are checked against the library. Every mode uses the same formatting normalization and two-attempt limit. An earlier parser diagnostic is retained separately. In the original three seeds, direct and structured complete 18/18 slots and raw 15/18; across all eight seeds, direct, raw and structured complete 48/48, 44/48 and 45/48. Random integrates $\binom{15}{6} = 5,005$ subsets; shared-first and realized-fit sensitivities are retained.

Table E.6: Fresh history-feedback comparison on the gene-disjoint, four-context cell benchmark (three paired training/controller seeds). This benchmark is separate from the original VCC, Norman and Tahoe tasks. Top-1 scores are percentages; $\pm$ reports the sample standard deviation across seeds. Search methods share the same 16 designs, four clients, development panels and six proposal slots; each valid candidate is fitted for 20 rounds. Fits include the initial-design fit. Random search integrates all size-six subsets of the 15 non-initial designs and selects using development evidence, not test scores. Raw history has three invalid slots across 18; direct and structured history complete all 18. Bold marks the best mean development or test score.

Search method	Development	Held-out	Fits
Fully trained fixed D00	22.40 ± 0.07	21.43 ± 0.10	1
Direct (no feedback)	22.68 ± 0.25	22.62 ± 0.21	7
Raw history	22.68 ± 0.25	22.62 ± 0.21	6 ± 1
Structured history	23.38 ± 0.48	22.89 ± 0.56	7
Random without replacement	23.06 ± 0.24	22.67 ± 0.21	7
GP–EI	$\mathbf{23.60 \pm 0.32}$	$\mathbf{23.15 \pm 0.35}$	7
Paired held-out contrasts (percentage points)			
Contrast	Mean difference	95% CI	W/T/L
Structured − Direct	$+0.263$	$[-1.022, 1.568]$	2/1/0
Structured − Random	$+0.220$	$[-0.290, 0.738]$	2/0/1
Structured − GP–EI	-0.264	$[-0.930, 0.413]$	0/2/1

W/T/L counts paired seeds. Confidence intervals use 10,000 paired perturbation-gene bootstrap resamples and describe conditional test-gene uncertainty for the fixed selected models, rather than uncertainty over additional training or controller runs.

Table E.7: Resource accounting for the fresh history comparison. Training columns are per-seed means; the last two columns total all three seeds. Fits are evaluated from a shared cache, with each policy charged for every candidate it requests. Raw-history invalid slots consume proposal budget without a candidate fit. Identical rounds, steps and examples do not imply identical FLOPs or language-model costs. Calls include permitted retries and logically charge the shared first proposal to each language-model arm.

Method	Rounds	Steps	Examples	Invalid slots	LLM calls
Fully trained fixed D00	20	2,420	76,120	0	0
Direct (no feedback)	140	16,940	532,840	0	18
Raw history	120	14,520	456,720	3	24
Structured history	140	16,940	532,840	0	21
Random without replacement	140	16,940	532,840	0	0
GP–EI	140	16,940	532,840	0	0

Original three-seed results. Structured feedback changes the selected design in two seeds, with two gains and one tie against direct. Raw and direct select identical final predictors. The positive mean contrasts in Table E.6 have gene-cluster intervals crossing zero, and GP–EI has the best mean. Intervals use 10,000 perturbation-gene resamples shared across contexts, methods and seeds, conditional on fixed selected predictors.

Eight-seed extension. Because three seeds leave run-to-run variation large relative to the contrasts, we added seeds 64–68 with the protocol, library, prompts, proposer and scorer unchanged. The analysis was registered before the new fits: the eight-seed mean and SD of every arm, paired structured-minus-direct, -random and -GP–EI contrasts with 100,000 seed-block bootstrap draws, and win/tie/loss counts, with no exclusions or replacement seeds. All selections were sealed before held-out scoring. Table E.8 reports every arm. Structured history beats direct proposals ($+0.41$ Top-1, 95% interval [0.09, 0.84]; four wins, four ties, no losses) and raw history ($+0.105$, [0.006, 0.280]), and exceeds fitted D00 in all eight seeds ($+1.38$). The registered primary contrast, structured history minus random search, is $+0.06$ ($[-0.33, 0.46]$; four wins, four losses), so history shows no clear advantage over random selection here; it trails GP–EI (-0.52, $[-0.94, -0.14]$). Direct search falls below fitted D00 once (seed 67). On this benchmark, development feedback therefore improves language-model proposals over feedback-free ones, while Bayesian optimization over the same 16 designs remains the strongest policy.

Table E.8: Eight-seed extension of the gene-disjoint cell comparison (Table E.6). Seeds 61–63 are the original runs; seeds 64–68 were added under an analysis fixed before their fits, with no replacement seeds or exclusions. Held-out Top-1 is in percent; $\pm$ is the sample SD over the eight seeds. Intervals resample the eight training/controller seeds as paired blocks (100,000 draws), so they describe run-to-run variation, unlike the test-gene intervals of Table E.6. Invalid slots over 48: raw 4, structured 3, direct 0; invalid requests consume their slots. Bold marks the best mean.

Search method	Held-out Top-1
Fully trained fixed D00	21.37 ± 0.22
Direct (no feedback)	22.34 ± 0.51
Raw history	22.65 ± 0.43
Structured history	22.75 ± 0.53
Random without replacement	22.69 ± 0.21
GP–EI	$\mathbf{23.28 \pm 0.34}$

Paired held-out contrasts (percentage points)

Contrast	Mean difference	95% CI	W/T/L
Structured − Direct	$+0.409$	$[0.088, 0.842]$	4/4/0
Structured − Raw	$+0.105$	$[0.006, 0.280]$	3/5/0
Structured − Random	$+0.061$	$[-0.325, 0.456]$	4/0/4
Structured − GP–EI	-0.524	$[-0.937, -0.142]$	1/3/4
Structured − Fixed D00	$+1.383$	$[0.984, 1.842]$	8/0/0

Original three-seed resource accounting. Each valid fit uses 2,420 optimizer steps and 76,120 example exposures; seven-fit searches use 16,940 steps and 532,840 exposures. Head parameter counts differ, so these are not equal-FLOP claims. Direct/raw/structured logically charge 18/24/21 model calls including retries; the shared first proposal gives 57 physical calls. Feature extraction and 48 fits cost about 0.236 GPU hours, excluding language-model inference and test scoring. All 48 held-out predictions are independently verified after selection.

Figure 6 retains every prefix, policy and seed of the original three seeds. All final searches in those seeds exceed fitted D00 (in the eight-seed extension, structured history exceeds it in every seed and direct search falls below it once), but seed 62 slot 2 raises development Top-1 by 0.057 points and lowers test Top-1 by 0.257 before recovery. K562 also retains a context-level reversal. Relative to the earlier benchmarks, the data, panels and library change together; within E.6, raw and structured modes compare feedback organization under matched conditions. None of these prefixes is used for test-selected stopping.

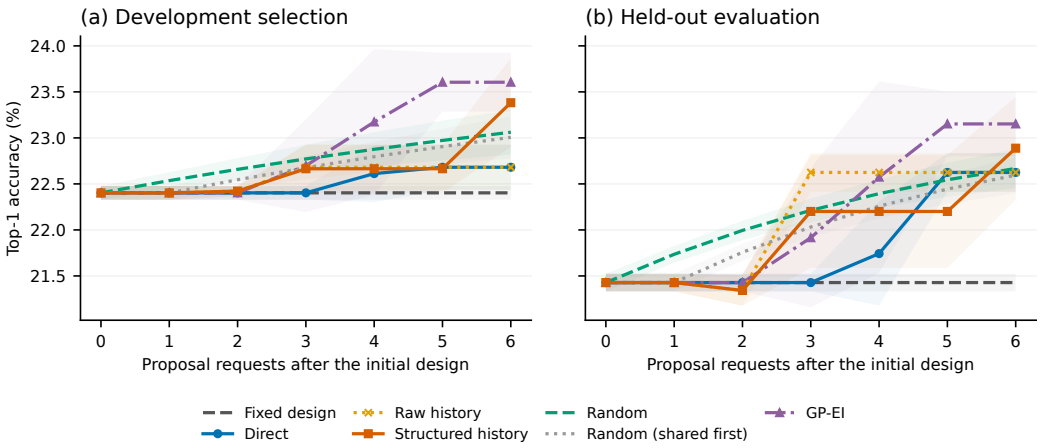


Figure 6: Every development and held-out prefix in the original three-seed gene-disjoint cell study (seeds 61–63), including zero and six requests. Lines are means over the three seeds; bands are sample SD. Random curves integrate subset choice within each seed; their error bars describe variation across seed-conditional expectations. Held-out curves score the development-selected model at each prefix.

E.7 Sensitivity to laboratory identity

Forty fresh CPU fits train D00 separately on laboratories 0–9 for protein, VCC, Norman and Tahoe. Every fit uses seed 42, 100 rounds, only its own training/development data and the common task test set after sealed selection. All 40 saved-prediction scores reproduce independently. SD here measures laboratory-identity variation, not training-seed variation.

Laboratory 0 is below the independent-laboratory mean for protein/VCC, near it for Tahoe, and best for Norman. This fixed-model sensitivity supplements the main GPU workflows; it does not rotate every controller. DTI rotation additionally requires laboratory-specific training dictionaries and a registered initialization protocol because the present dictionary derives from laboratory 0.

Table E.9: Fixed-design sensitivity to the identity of the single participating laboratory. Each of ten laboratories independently fits D00 for 100 rounds and selects a checkpoint on its own development panel (seed 42). All 40 fits are fresh CPU runs under the same training equations; the SD and range describe laboratory identities, not training seeds. This diagnostic does not rerun research controllers.

Endpoint	Lab 0	Single-lab mean $\pm$ SD	Range
Proteomics AP	24.01	28.69 ± 6.44	14.69–37.34
VCC Top-1	29.42	32.50 ± 3.49	26.96–39.76
Norman Top-1	25.56	17.67 ± 5.84	4.37–25.56
Tahoe Top-1	19.40	19.70 ± 4.66	13.43–26.87

E.8 Repaired panel feedback and the confirmation gap

A versioned source-defined adapter/reducer/projection retains each target panel's identifier, primary score and loss, distinguishing four training laboratories, ten development panels and one evaluation worker. Worst-panel score and SD now use the actual vector; the primary/loss selection key is unchanged. Raw/structured receive identical observed facts and direct no scores. Unsaved old intermediate vectors remain unavailable.

A frozen audit reconstructs all twelve designs at seeds 61–63 on both tasks: 72 checkpoints and 720 labeled panel scores. Primary/loss reproduce exactly; true worst differs from mean in every checkpoint. Eight paired ten-round CPU canaries on D00/D07 preserve training losses, development scores, selected rounds/weights and final states. These verify implementation, not a search-performance gain.

Table E.10: Development-only audit after restoring target-panel feedback. Each row reconstructs all twelve saved best checkpoints. Gap is the mean across designs of primary minus the true worst-panel score, in percentage points. A/B are fixed even/odd panel subsets. The last column counts five-panel proposal subsets selecting the same design as all ten panels, out of 252. Complement scores reuse checkpoints already selected on all ten panels; they are descriptive, not independent confirmation.

Task	Seed	Mean–worst gap	A gain	B gain	Winner agreement
Cell	61	20.13	+17.13	-2.68	102/252
Cell	62	19.77	+13.58	-8.54	77/252
Cell	63	19.67	+13.18	+13.19	166/252
Protein	61	54.63	+0.00	+0.00	43/252
Protein	62	53.93	+0.00	+0.00	16/252
Protein	63	55.20	+1.67	+0.00	72/252

Even-indexed panels select among saved checkpoints and odd-indexed panels provide a complementary diagnostic. Cell seeds 61/62 improve the first subset by 17.13/13.58 points but lose 2.68/8.54 on its complement; seed 63 improves both. Over all 252 five-panel proposal subsets, the all-panel winner survives in 102/77/166 cell and 43/16/72 protein cases. Subsets overlap, loss breaks primary ties, and checkpoints were originally selected on all panels, so these complementary-panel results are descriptive; they verify feedback transport.

E.9 WIDER RECIPE CHOICES AT MATCHED ACCESS AND CANDIDATE BUDGETS

The expanded library trains 36 recipes at seeds 71–73 on DTI and observed-response proteomics: 216 new fits shared by all search policies. The recipe library and selection protocol are fixed before new test evaluation; the 110-fit development pilot is reported separately in the construction cost.

Models, recipes and development selection. Every policy starts from the same original fixed-default seed42 predictor per task. DTI freezes the fitted TAPB encoder and adapts its per-confounder classifier while retaining the original confounder-prior probability mixture. Protein either updates the complete original observed-response Conv1d/MLP head, or freezes it and adds a residual from the original 12,390 protein-response and Morgan-fingerprint inputs. These are warm-start studies, not replacements for the original end-to-end scores. The new seeds vary subsequent training and proposal generation, conditional on one common historical starting model; they are not three independent backbone fits.

The full recipe product combines learning rate $\{10^{-5}, 10^{-4}, 10^{-3}\}$, original-head fitting or a zero-output-initialized 32/128-wide residual, BCE or BCE plus 0.1 times within-batch positive–negative softplus ranking loss, and constant or cosine learning-rate schedules. The cosine schedule ends at 0.1 of its initial rate. Every fit completes 20 full-client rounds with AdamW reset locally each round, weight decay 0.001, gradient clipping at one and sample-weighted aggregation. Batch sizes are 256 for DTI and 64 for protein. Checkpoints at rounds 2/5/10/20 are selected by pooled development AUROC/AP, then lower pooled BCE and earlier round. This pooled selection differs from the original equal-panel protocol; all methods change together. Ten-panel diagnostics remain available, with undefined one-class metrics marked explicitly.

Matched proposers and conventional controls. Qwen2.5-7B and the Luna service each make six requests after a fitted D00, with one real first proposal shared across direct/raw/structured modes. All modes use the same recipe library, initialization, checkpoint rule, two-attempt limit and failure charges. Raw/structured receive the same queried facts, including all recorded panel trajectories. A reversible panel-table encoding prevents history truncation; every request checks exact reconstruction of its original card. Earlier interrupted controller runs remain engineering records. The Luna transport accepts only a completed text turn with recorded usage and no tool actions. Its generation seed and output-token cap cannot be enforced, and CLI context overhead is included in its token counts.

Random search exactly integrates all $\binom{35}{6} = 1{,}623{,}160$ subsets and returns the expectation of a development-selected single predictor. These subsets are not independent training repetitions or prediction ensembles. We retain GP–EI with the original offset $\xi = .01$ and add $\xi = 0$ as a stronger small-score-scale control, both at six requests. Shared-first and realized-fit-count random sensitivities are retained. Both proposer comparisons reuse the same control fits; repeated control values are not additional replications.

Table E.11: Expanded-recipe comparison on DTI and proteomics. Each model uses the same 36 designs, ten training laboratories, six requests and 20 rounds per valid fit. Scores are means ± sample SD over three new head-training/controller runs, conditional on one shared seed42 starting predictor per task. DTI freezes the prior TAPB encoder; protein refits the complete observed-response head or a residual from all original inputs. Both proposer models and both GP offsets are retained. Bold marks the best means, including ties, not significance. These retrospective warm-start studies are separate from the original end-to-end scores.

	Qwen2.5-7B		Luna service	
Policy	DTI AUROC	Protein AP	DTI AUROC	Protein AP
Fully fitted D00	94.271 ± 0.000	37.109 ± 0.000	94.271 ± 0.000	37.109 ± 0.000
Direct	94.334 ± 0.054	38.244 ± 0.465	$\mathbf{94.368 \pm 0.006}$	$\mathbf{39.017 \pm 0.659}$
Raw history	$\mathbf{94.368 \pm 0.006}$	38.533 ± 0.886	$\mathbf{94.368 \pm 0.006}$	38.572 ± 0.558
Structured history	94.351 ± 0.036	38.616 ± 0.826	$\mathbf{94.368 \pm 0.006}$	38.572 ± 0.558
Random expectation	94.316 ± 0.007	38.655 ± 0.278	94.316 ± 0.007	38.655 ± 0.278
GP–EI ($\xi = .01$)	94.290 ± 0.029	$\mathbf{39.017 \pm 0.659}$	94.290 ± 0.029	$\mathbf{39.017 \pm 0.659}$
GP–EI ($\xi = 0$)	94.336 ± 0.056	37.975 ± 0.837	94.336 ± 0.056	37.975 ± 0.837
Structured minus random; percentage points and conditional 95% intervals				
Proposer	DTI Δ	DTI interval	Protein Δ	Protein interval
Qwen	$+0.035$	$[0.018, 0.054]$	-0.040	$[-1.616, 1.579]$
Luna	$+0.053$	$[0.031, 0.077]$	-0.083	$[-2.025, 2.234]$

Original three-seed outcomes and scope. Luna structured history reaches DTI AUROC 94.368 versus random's 94.316, a $+0.053$-point difference with paired conditional 95% interval [0.031, 0.077] and gains in all three runs. Direct and raw history select the same DTI predictors. On protein, direct reaches 39.017 AP versus random's 38.655, while structured/raw history reaches 38.572. The prespecified structured-minus-random protein contrast is -0.083 points, with interval $[-2.025, 2.234]$; Qwen's is -0.040 $[-1.616, 1.579]$. Protein direct and the original GP–EI select identical final predictors in each seed. Thus the expanded workflow supplies models above random on both tasks, but the primary history comparison does not win both. This result does not establish an advantage from structured history over raw history or conventional search.

All twelve task/seed/proposer selections are sealed before this study's test evaluation; saved predictions independently reproduce all 216 scores. Intervals use 2,000 paired bootstrap draws over DTI rows or protein compounds, shared across methods and seeds. They describe fixed-predictor test-sample uncertainty, not additional controller/backbone runs. The tests were exposed previously: this retrospective follow-up provides no new biological-split evidence. The separate cell benchmark in E.6 adds no new recipe-study result.

Eight-seed extension. As for E.6, we added seeds 64–68 (360 new fits) with the recipe library, starting predictors, prompts, proposers, checkpoint rule and scorer unchanged. The analysis was registered before the new fits: the eight-seed mean and SD of every arm; the primary Luna structured-minus-random contrast for each task; structured-minus-direct, -raw and -GP–EI contrasts; 100,000 seed-block bootstrap draws; and win/tie/loss counts, with no exclusions or replacement seeds. All selections were sealed before held-out scoring, and the original seeds reproduce Table E.11 exactly. Table E.12 reports every arm. On DTI, structured history exceeds random search for both proposers (Luna $+0.053$ AUROC, 95% interval [0.050, 0.055], all eight seeds; Qwen $+0.040$, [0.021, 0.054], seven of eight) and exceeds the original GP–EI in every seed; Luna's direct, raw and structured modes select the same predictor in every seed. On protein, the primary Luna contrast is $+0.21$ AP ($[-0.04, 0.44]$; six wins, two losses), compared with -0.08 over the original three seeds; Luna direct search (38.99) and GP–EI (39.02) have higher means, and Qwen structured history is below random (-0.44, $[-0.96, 0.09]$). Every structured-history search exceeds the fitted D00 starting design in all eight seeds of all four proposer–task columns. Qwen's invalid requests (30–36 of 96 per mode) consume their slots; Luna completes every slot. With wider recipes, language-model search therefore stays above random on DTI, while on protein Bayesian search and feedback-free proposals are at least as strong as history.

All-valid searches use seven complete fits and 140 full-client rounds: 28,000 optimizer steps/7,020,860 examples for DTI and 1,400/54,040 for protein. Different heads still have differ-

Table E.12: Eight-seed extension of the expanded-recipe comparison (Table E.11). The original three head-training/controller runs are retained; five runs were added under an analysis fixed before their fits, with no replacement seeds or exclusions. Each run keeps the same 36 designs, ten training laboratories, six requests, 20-round fits and the shared seed-42 starting predictor per task. Scores are percentages; ± is the sample SD over the eight runs. Contrasts give the mean paired difference in percentage points, a seed-block bootstrap 95% interval (100,000 draws) and wins/ties/losses over the eight runs. Invalid requests over 96 per proposer and arm: Qwen direct/raw/structured 36/36/30; Luna direct/raw/structured 0/0/0; they consume their slots. Bold marks the best mean per column, including ties.

Policy	Qwen DTI	Qwen protein	Luna DTI	Luna protein
Fully fitted D00	94.271 ± 0.000	37.109 ± 0.000	94.271 ± 0.000	37.109 ± 0.000
Direct	94.359 ± 0.036	38.611 ± 0.745	$\mathbf{94.372 \pm 0.005}$	38.994 ± 0.537
Raw history	$\mathbf{94.372 \pm 0.005}$	38.466 ± 0.944	$\mathbf{94.372 \pm 0.005}$	38.751 ± 0.703
Structured history	94.359 ± 0.026	38.142 ± 0.724	$\mathbf{94.372 \pm 0.005}$	38.789 ± 0.644
Random expectation	94.319 ± 0.006	38.581 ± 0.381	94.319 ± 0.006	38.581 ± 0.381
GP–EI ($\xi = .01$)	94.286 ± 0.021	$\mathbf{39.016 \pm 0.544}$	94.286 ± 0.021	$\mathbf{39.016 \pm 0.544}$
GP–EI ($\xi = 0$)	94.339 ± 0.048	38.440 ± 0.781	94.339 ± 0.048	38.440 ± 0.781
Paired contrasts (percentage points), 95% interval, W/T/L				
Structured − Direct	$+0.000$ (1/5/2)	-0.469 (2/1/5)	$+0.000$ (0/8/0)	-0.205 (2/1/5)
	$[-0.03, 0.03]$	$[-1.27, 0.17]$	$[0.00, 0.00]$	$[-0.42, -0.01]$
Structured − Raw	-0.013 (0/6/2)	-0.325 (3/4/1)	$+0.000$ (0/8/0)	$+0.038$ (1/7/0)
	$[-0.03, 0.00]$	$[-1.08, 0.09]$	$[0.00, 0.00]$	$[0.00, 0.12]$
Structured − Random	$+0.040$ (7/0/1)	-0.439 (2/0/6)	$+0.053$ (8/0/0)	$+0.209$ (6/0/2)
	$[0.02, 0.05]$	$[-0.96, 0.09]$	$[0.05, 0.06]$	$[-0.04, 0.44]$
Structured − GP–EI ($\xi = .01$)	$+0.074$ (8/0/0)	-0.874 (0/2/6)	$+0.086$ (8/0/0)	-0.227 (1/2/5)
	$[0.05, 0.09]$	$[-1.41, -0.41]$	$[0.07, 0.10]$	$[-0.42, -0.06]$
Structured − GP–EI ($\xi = 0$)	$+0.020$ (4/2/2)	-0.298 (4/0/4)	$+0.032$ (4/4/0)	$+0.350$ (3/3/2)
	$[-0.02, 0.06]$	$[-1.05, 0.36]$	$[0.01, 0.06]$	$[-0.01, 0.85]$

Table E.13: Actual candidate-cache construction and logical search budgets. Each task has 108 new fits. A fully valid six-request search uses seven fits. Different heads have different parameter counts and FLOPs; optimizer steps and example exposures are matched. Historical initial-predictor fitting, feature extraction, the 110-fit development pilot and interrupted engineering controller runs are reported separately in the provenance.

Task	Fits	Fit seconds	Steps/search	Exposures/search
DTI	108	1714.7	28,000	7,020,860
Protein	108	115.6	1,400	54,040

ent FLOPs and runtime. Luna completes every logical slot; Qwen failures remain charged, with lower realized-fit random controls provided separately. Table E.13 reports the original three seeds (71–73); the eight-seed extension adds 180 fits per task, for 576 in total. The original three-seed proposer runs use 182 Qwen and 101 Luna physical completions, including repairs. Common historical fitting, shared feature extraction, the pilot and interrupted engineering calls are additional costs, separate from each search's logical budget.